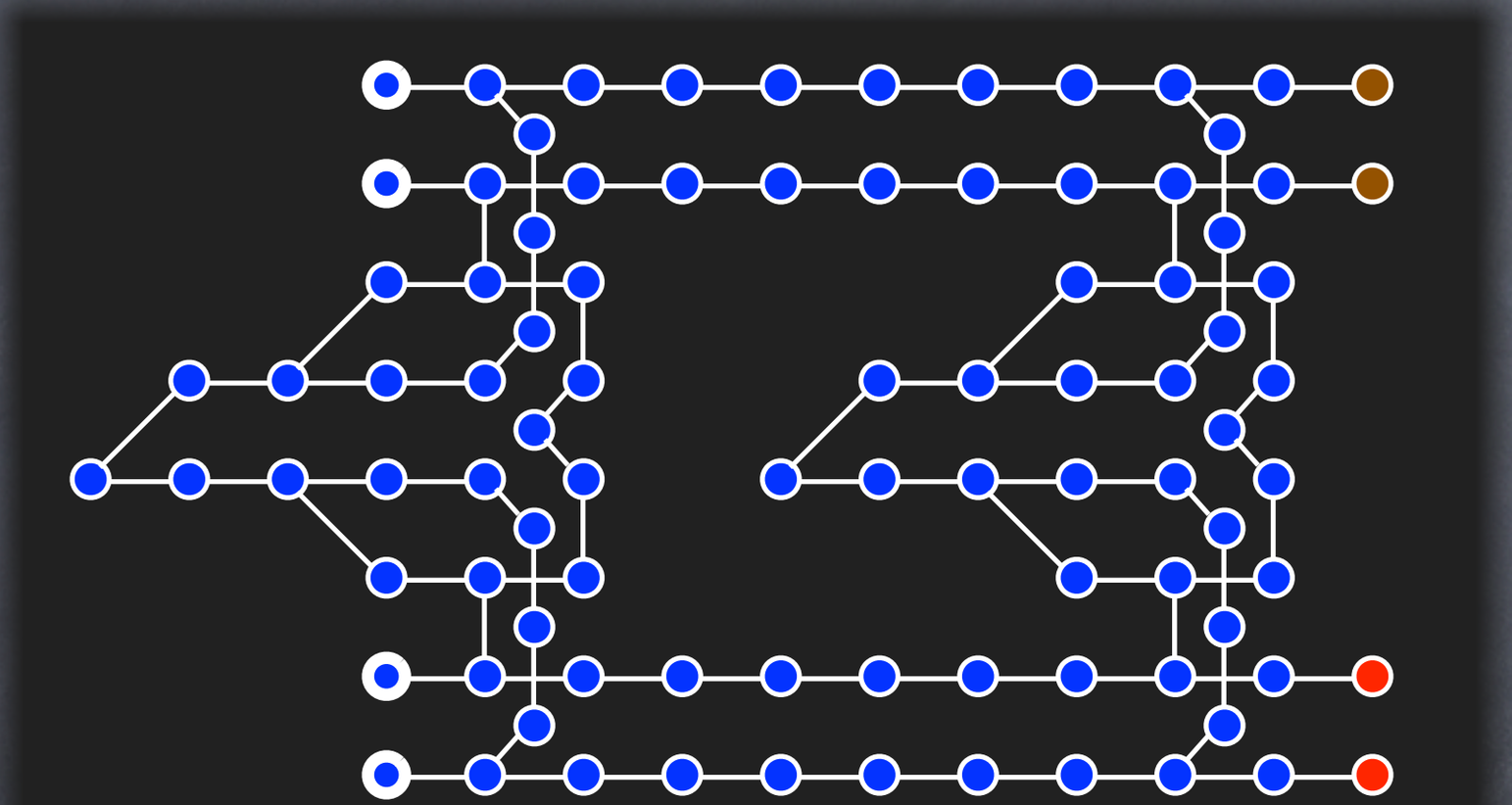
# Making Error Correction Spatially Local



Dave Bacon
University of Washington

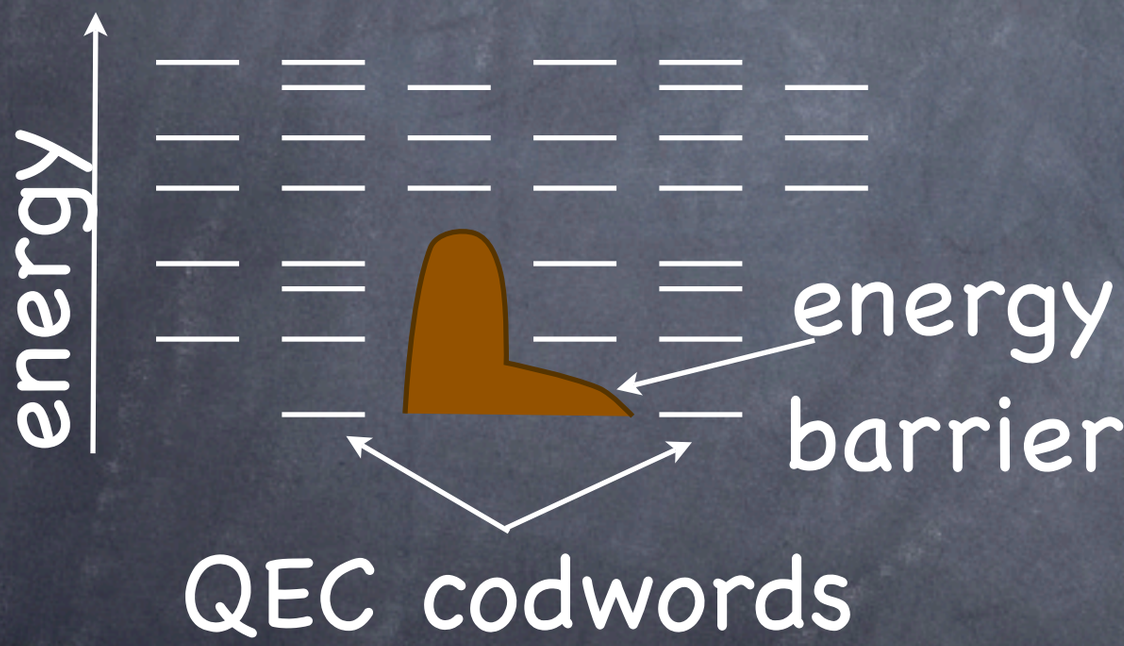
DARPA QuEST Program Review

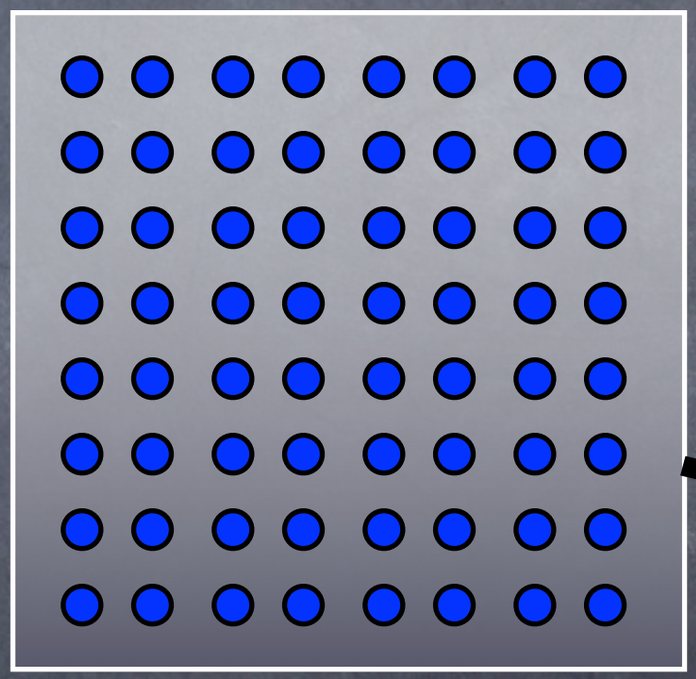
# First QuEST

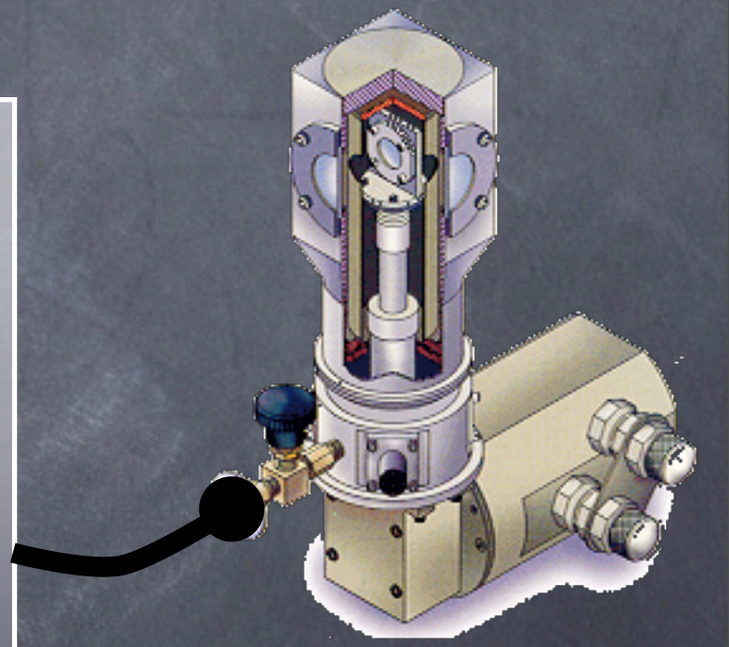


Do self-correcting quantum systems exist in realistic settings (spatially local few body interactions in 2 or 3 dimensions)?

**self correcting**



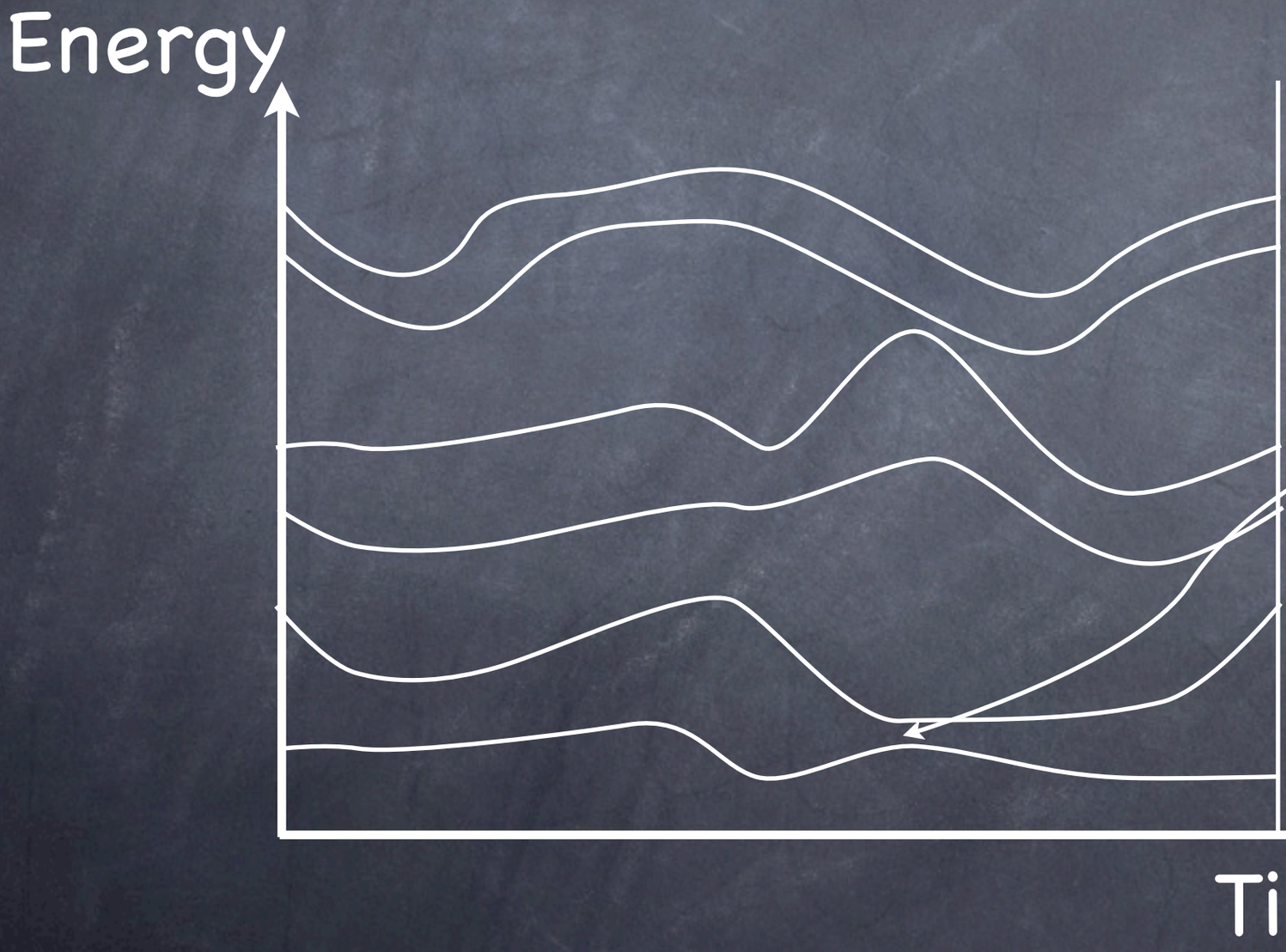


dream: quantum error correction is performed by simply keeping the many-body system cold

# Second QuEST



Is universal adiabatic quantum computing fault-tolerant?




polynomially shrinking energy gap

$$\frac{1}{\sqrt{T}} \sum_{t=0}^{T-1} |t\rangle |\psi(t)\rangle$$

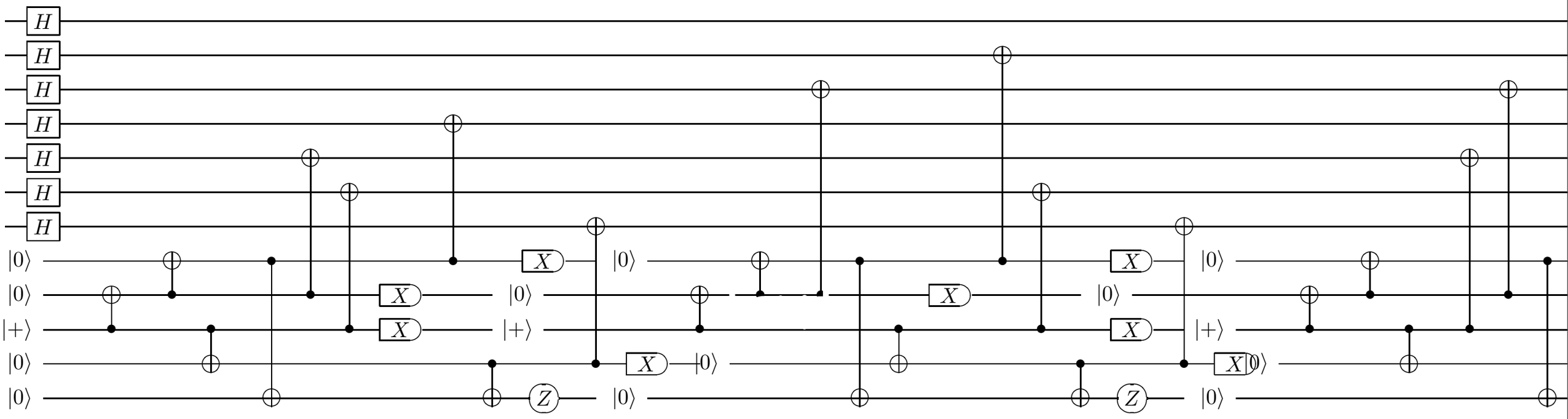
# Today

A result that has the potential
to impact both QuESTs

+

Bonus: may help in the experimental
implementation of quantum error correction

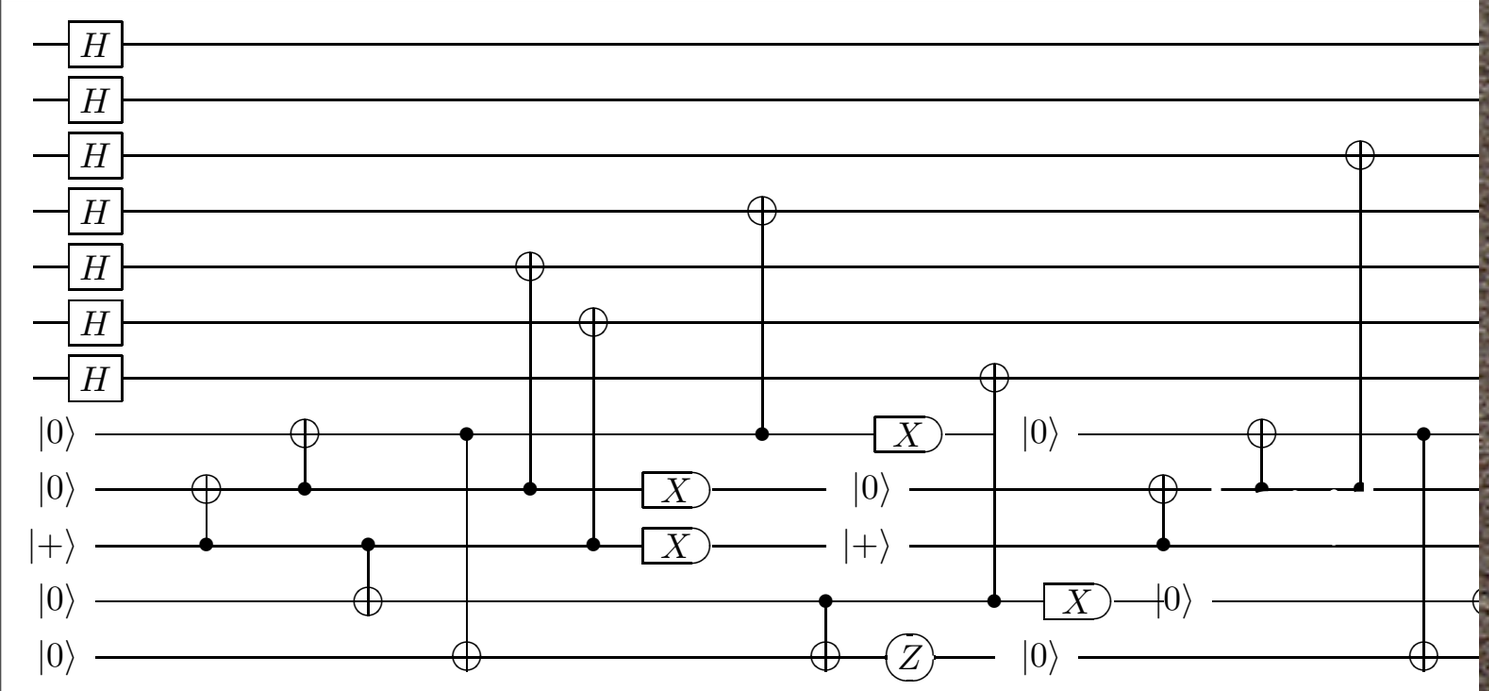
# Quantum Error Correction




Time ⟶

Complicated, time dependent many-qubit recipe for protecting quantum information

# Quantum Error




Time —

Complicated, time dependent

recipe for protecting qubits

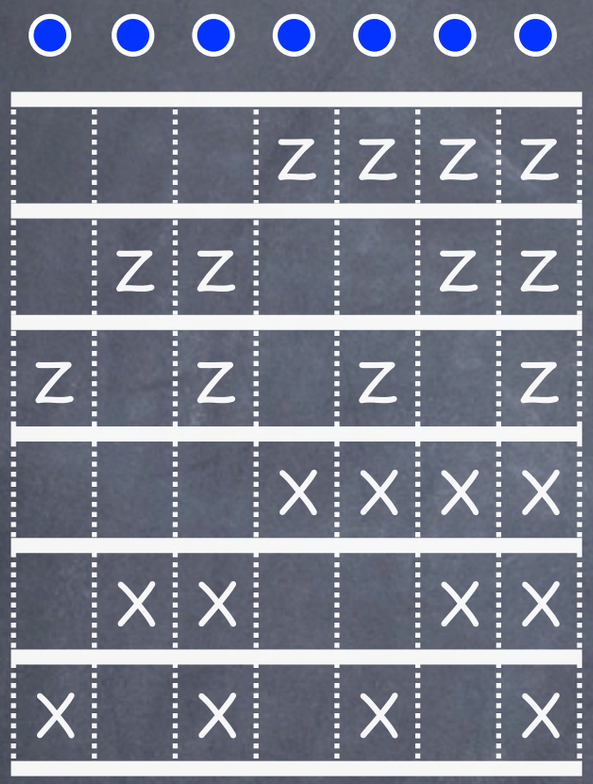
I CAN'T FIT N EXPERIMENTALIST'S BOX?

LOCALITY???



# Stabilizer Codes

[[7,1,3]] Steane code  QEC Recipe



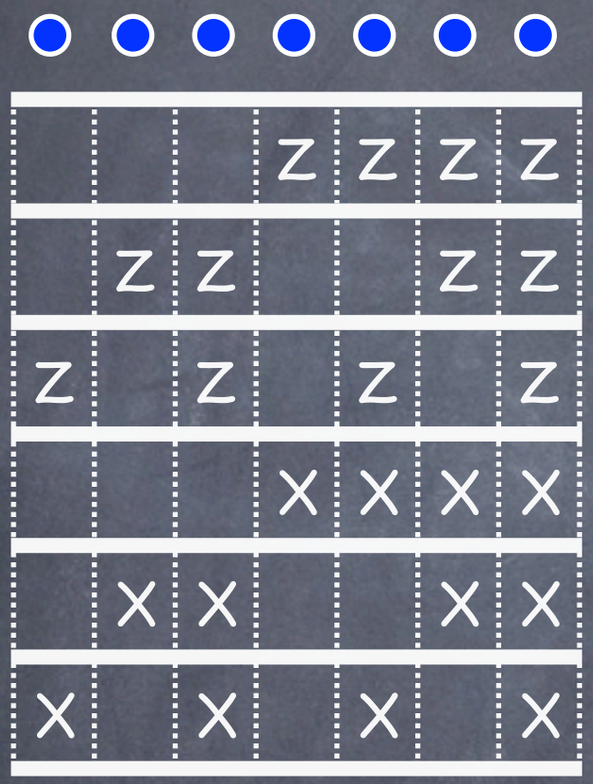

row = stabilizer

column = qubit

[[# qubits, # encoded qubits, distance]]

# Stabilizer Codes

[[7,1,3]] Steane code




row = stabilizer

column = qubit

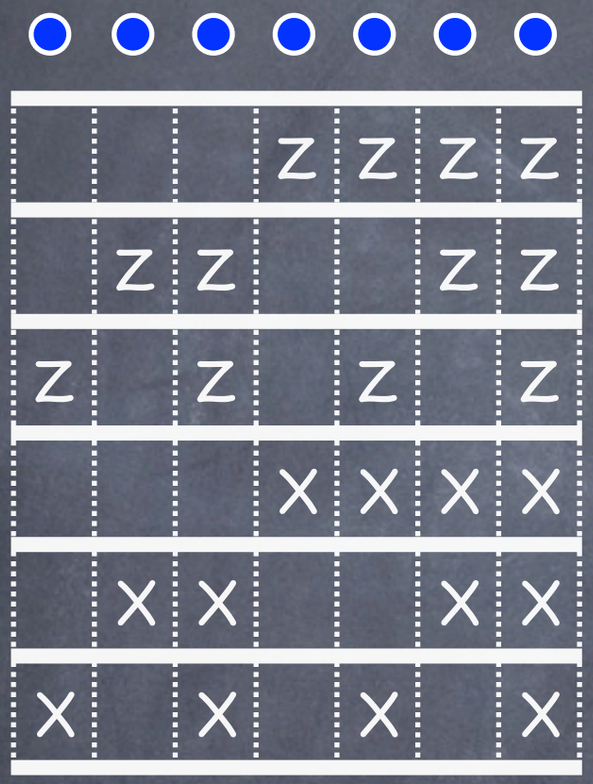
QEC Recipe

1. Measure 6 stabilizer operators

[[# qubits, # encoded qubits, distance]]

# Stabilizer Codes

[[7,1,3]] Steane code




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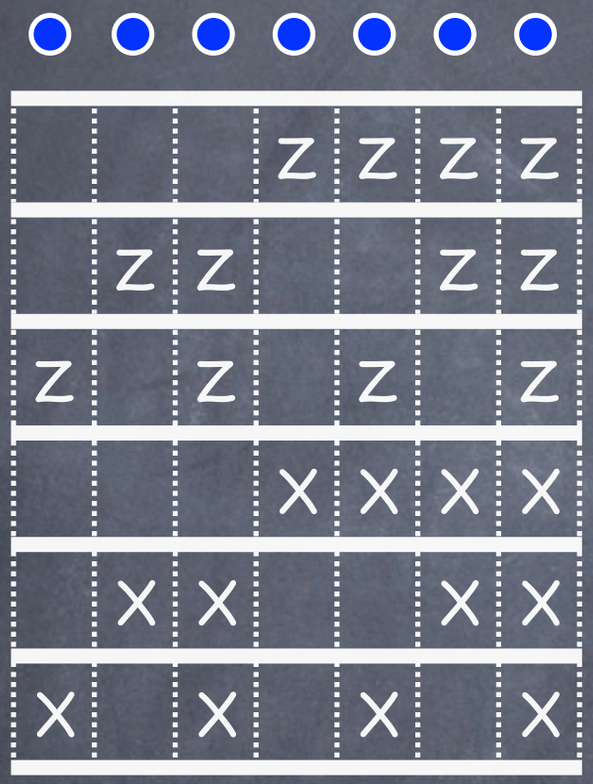
QEC Recipe

1. Measure 6 stabilizer operators
2. Conditional on measurement diagnose (and fix) single qubit error

[[# qubits, # encoded qubits, distance]]

# Stabilizer Codes

[[7,1,3]] Steane code

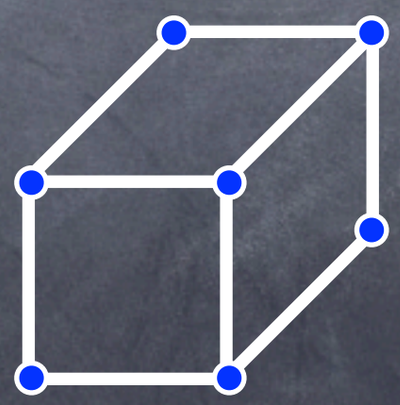



row = stabilizer

column = qubit

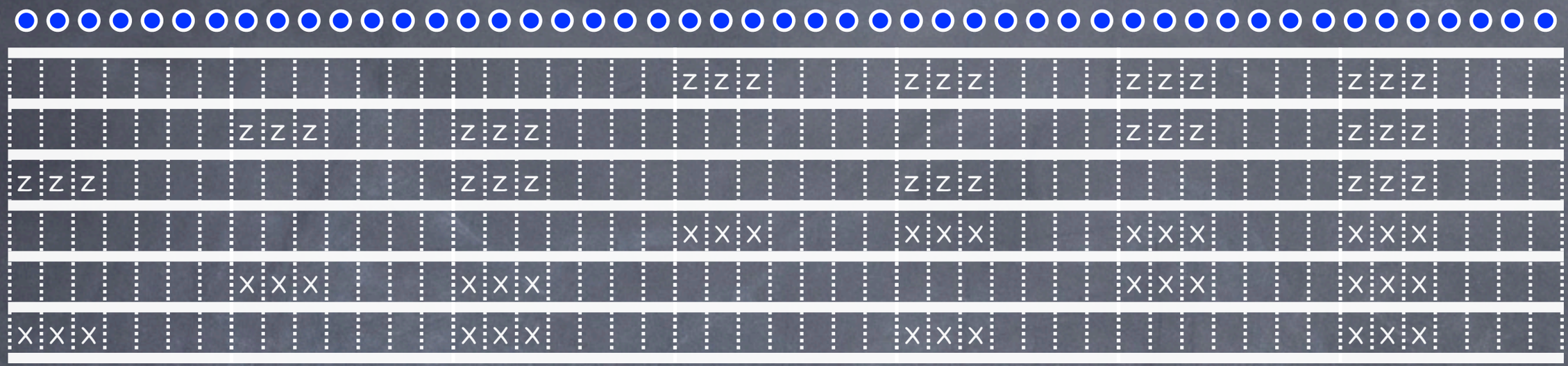
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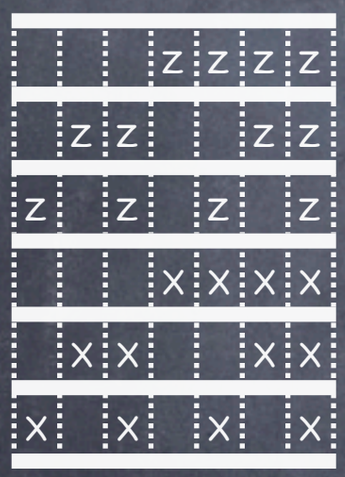
1. Measure 6 stabilizer operators
2. Conditional on measurement diagnose (and fix) single qubit error



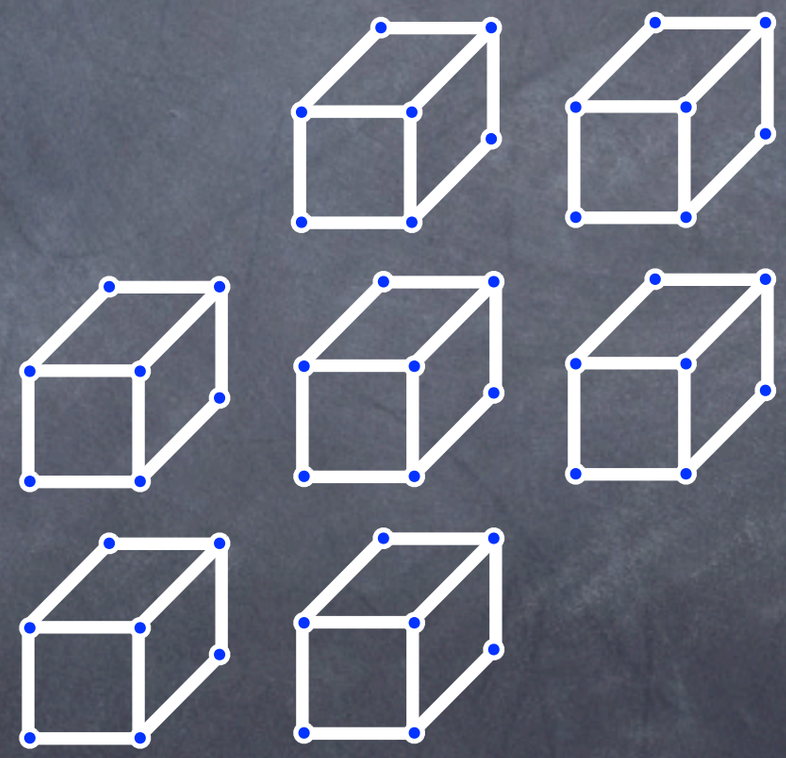
[[# qubits, # encoded qubits, distance]]

# Concatenating Our Lives Away





×6



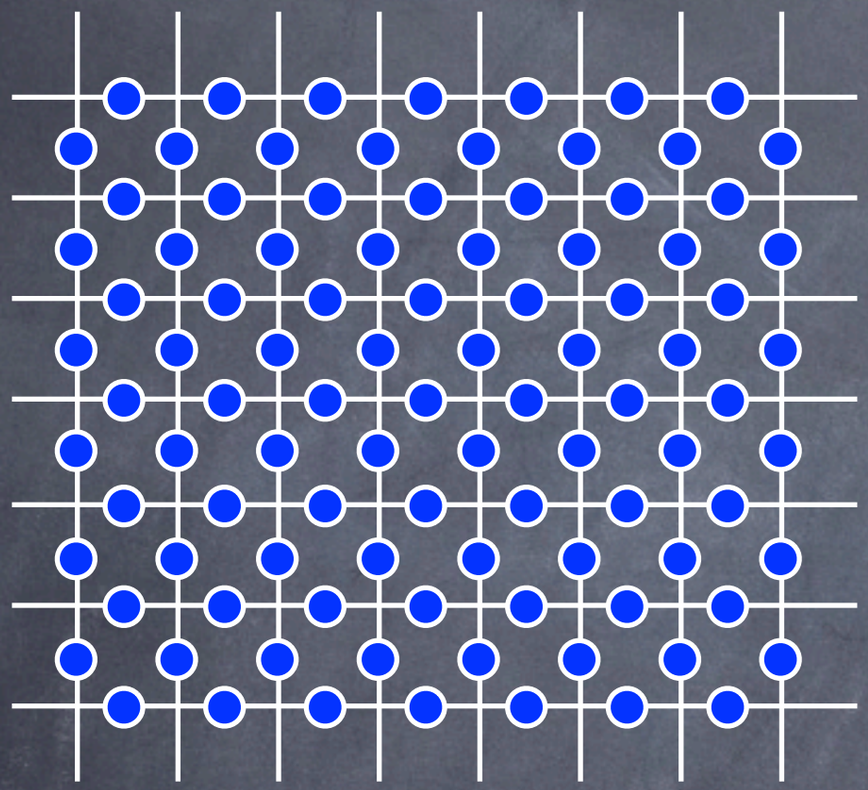
Concatenation implies measurement of spatially distant, high weight, Pauli operators

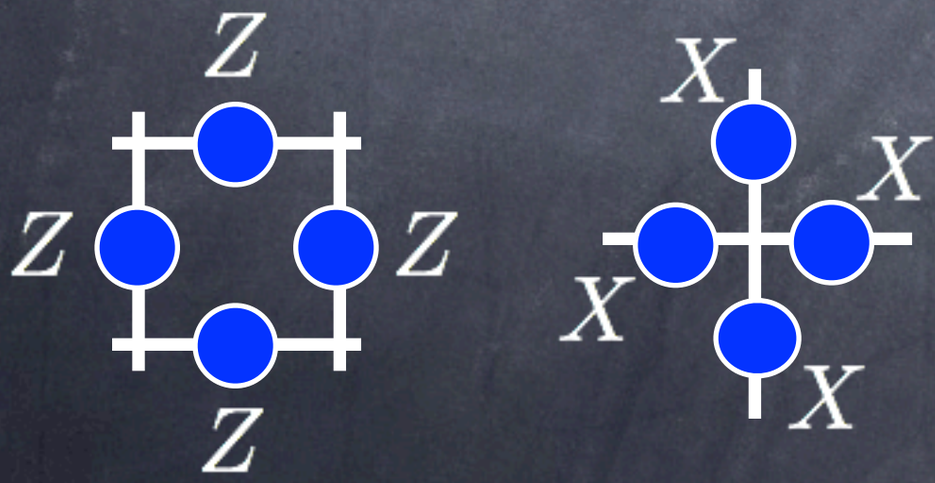
# Conca es Away




Concatena spatially distant, hig

# Rays of Hope





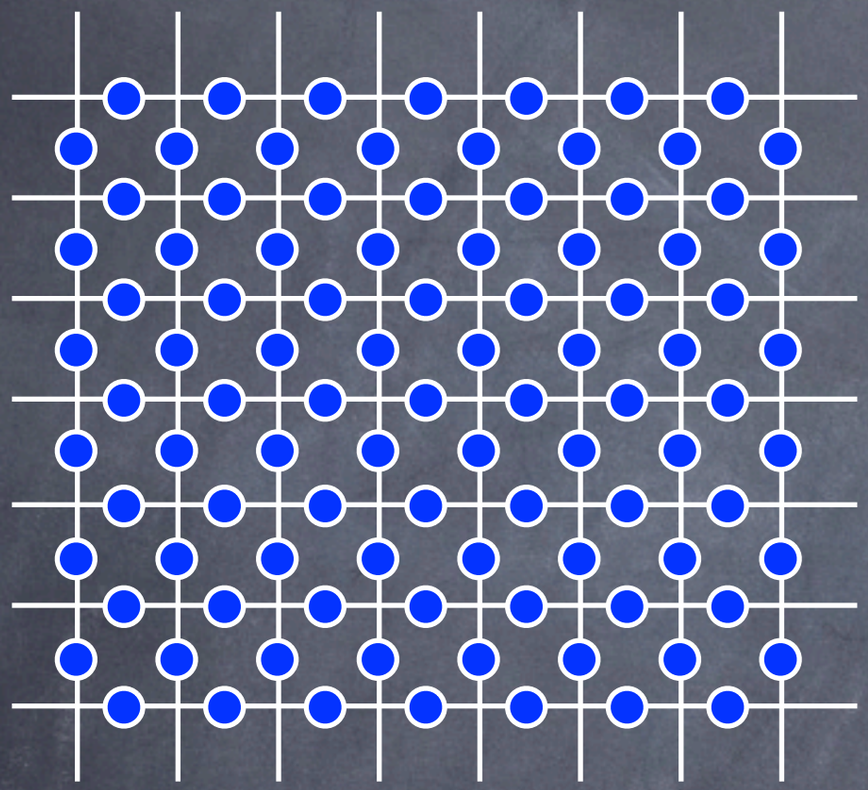

plaquette vertex

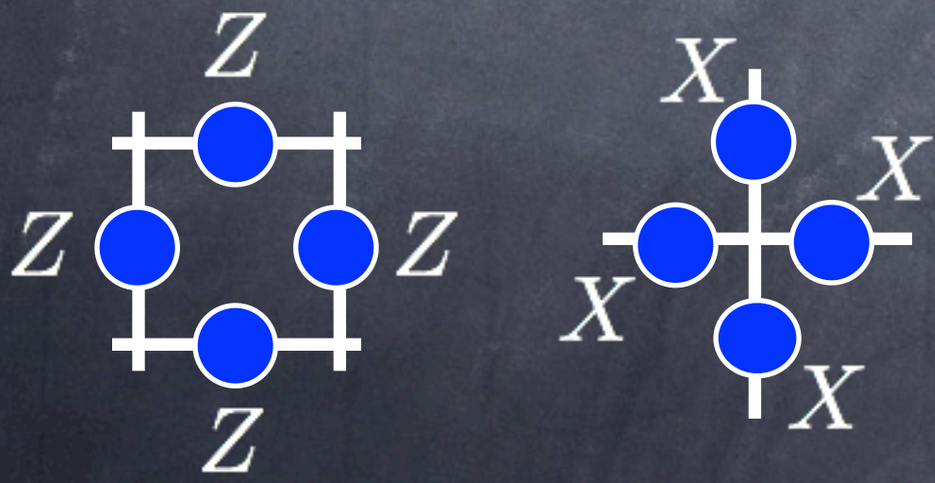
## Surface code QEC

1. Measure plaquette and vertex operators

2. Classically post-process measurement results to diagnose error

# Rays of Hope






plaquette vertex

## Surface code QEC

1. Measure plaquette and vertex operators

2. Classically post-process measurement results to diagnose error

# t-Local and t-Neighboring

weight of Pauli operator = # of operators ≠ I

➡ I⊗X⊗X⊗Y⊗Y⊗I⊗Z⊗I has weight 5

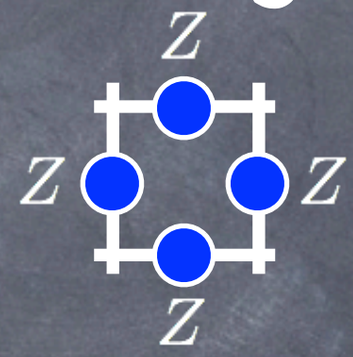
# t-Local and t-Neighboring

weight of Pauli operator = # of operators ≠ I

➡ I⊗X⊗X⊗Y⊗Y⊗I⊗Z⊗I has weight 5

t-Local code: exists a set of stabilizer generators whose weight ≤ t

➡ surface code is 4-local



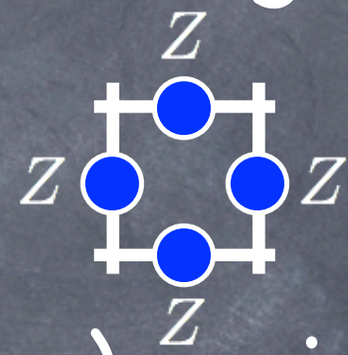
# t-Local and t-Neighboring

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t-Local code: exists a set of stabilizer generators whose weight ≤ t

➡ surface code is 4-local




t-Neighboring code (in D dimensions): exists a set of stabilizer generators where all ≠ I operators are within a distance t of each other on some lattice of qubits in D spatial dimensions

➡ surface code is 2-neighboring in 2D

# Lament



Why can't all codes be k-neighboring in two or three spatial dimensions?

# Main Result

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using t-neighboring measurements (t constant).

Every stabilizer code can be made to act like the surface code!

n = number of physical qubits
k = number of encoded qubits
d = distance of code
r = number of gauge qubits

# Main Result

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndr n be made using t-neig (t constant).

Every s nade to act

n = number o
k = number o
d = distance
r = number of gauge qubits

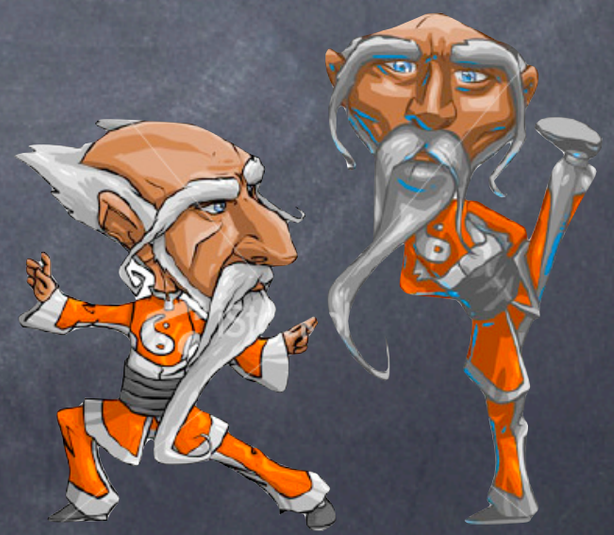


# Main Result

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using t-neighboring measurements (t constant).

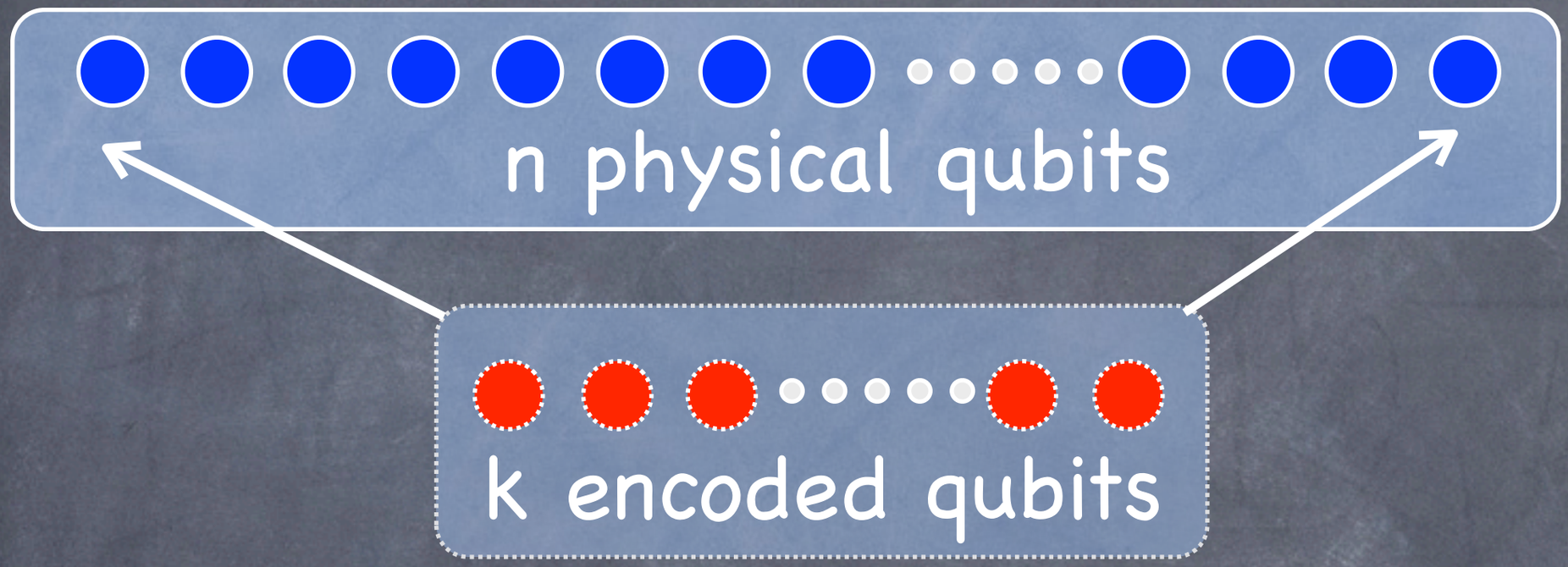
To grok, master these steps:

1. stabilizer subsystem codes
2. spaced-time locality in QEC
3. measurement based quantum computing



# Stabilizer Subsystem Code

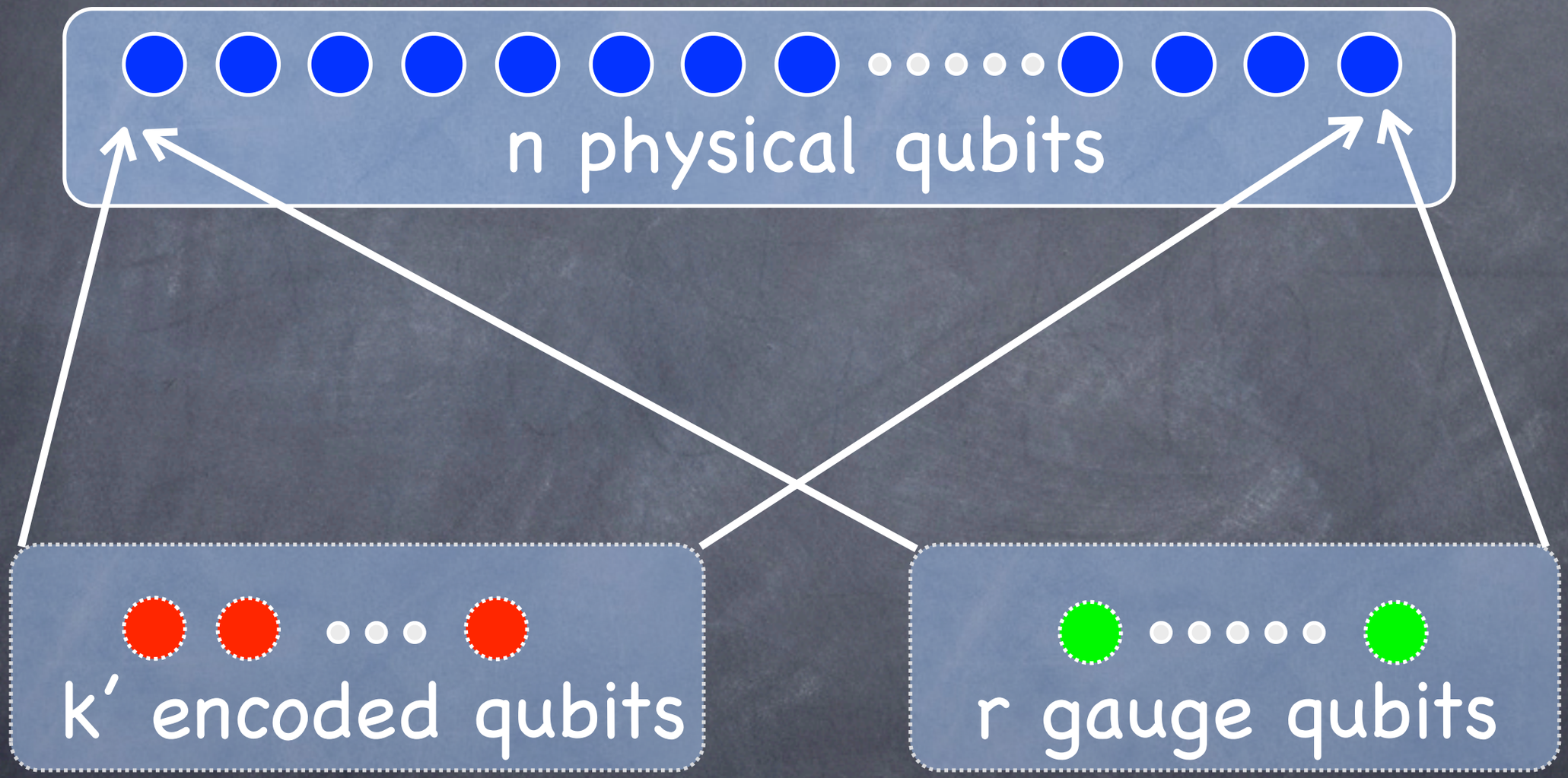
[n,k,d] Stabilizer Subspace Code




[n,k',r,d] Stabilizer Subsystem Code
[David Poulin, arXiv:quant-ph/0508131]

# Stabilizer Subsystem Code

[n,k,d] Stabilizer Subspace Code



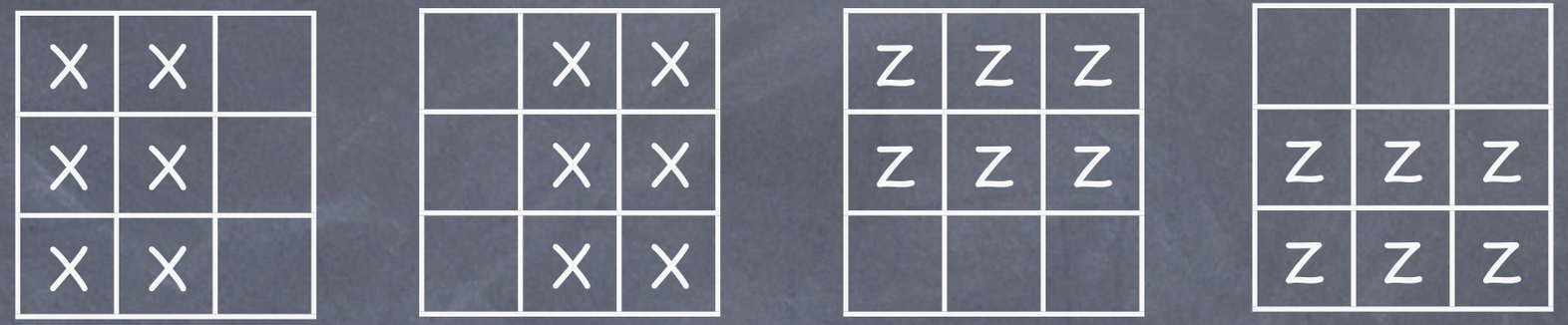

[n,k′,r,d] Stabilizer Subsystem Code

[David Poulin, arXiv:quant-ph/0508131]

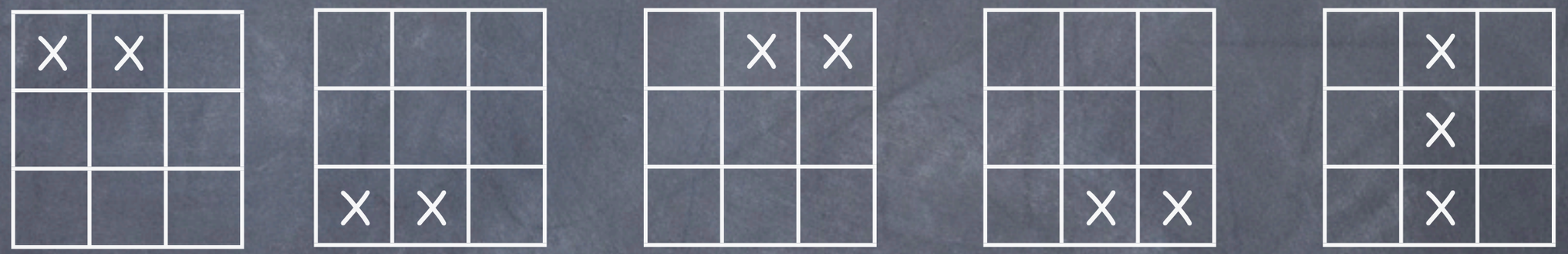
# Baloney Sandwich Code

[Bacon, arXiv:quant-ph/0506023]

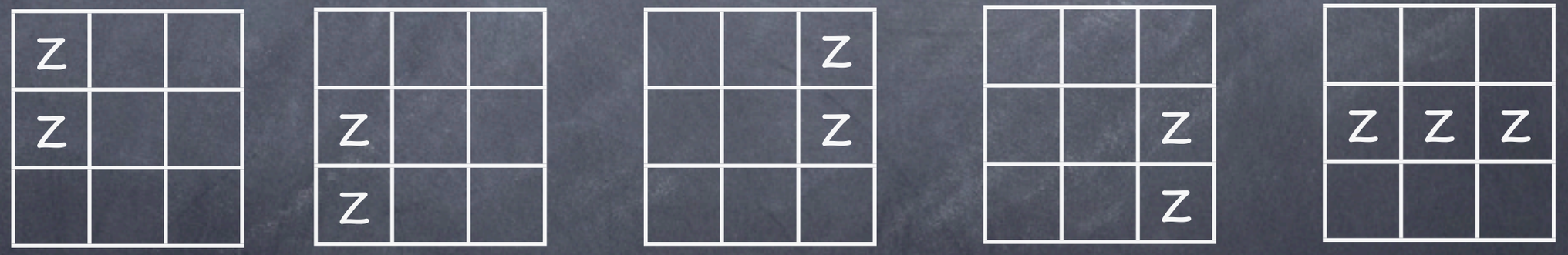
stabilizer generators




logical X




logical Z



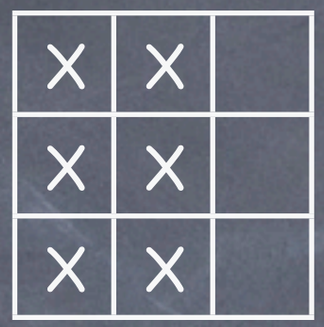

gauge

encoded

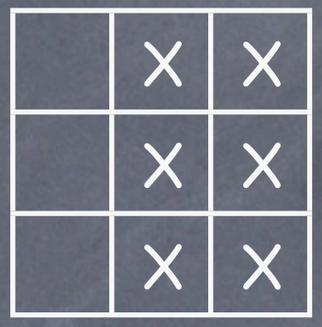
[[9,1,4,3]] stabilizer subsystem code

# Baloney Sandwich Code

stabilizer generators {



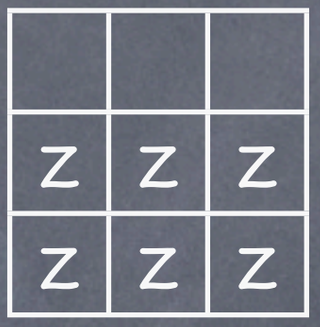
$S_1$



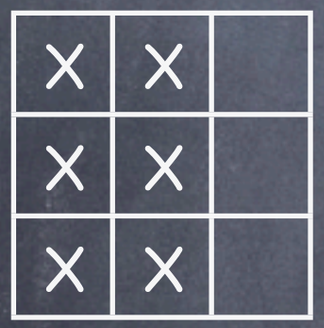
$S_2$

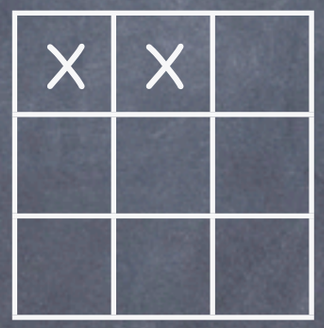


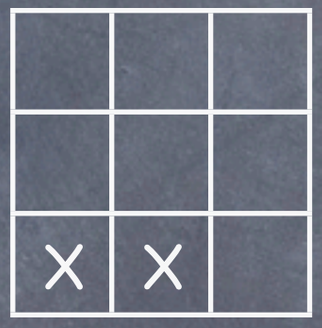
$S_3$



$S_4$



$S_1$ = 

$X_1$ X 

$X_2$ X 

$X_1X_2S_1$

We can measure $S_1$ by measuring 2-local operators that are products of gauge and stabilizer operators

6-local → 2-local

# Space-Time Neighboring

Any quantum circuit can be made to be neighboring in space-time for any spatial dimension D

Example in 1D



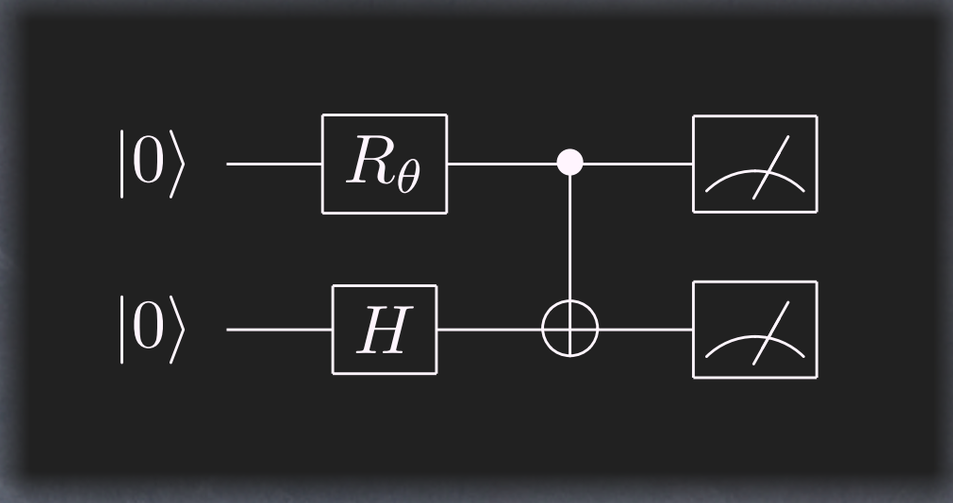
# Space-Time QEC

Fault-tolerant quantum error correction can be done using t-neighboring locality in space-time while maintaining a threshold.

(1D requires next-nearest neighbor gates)

[Aharonov, Ben-Or, arXiv:quant-ph/9906129]
[Gottesman, arXiv:quant-ph/9903099]

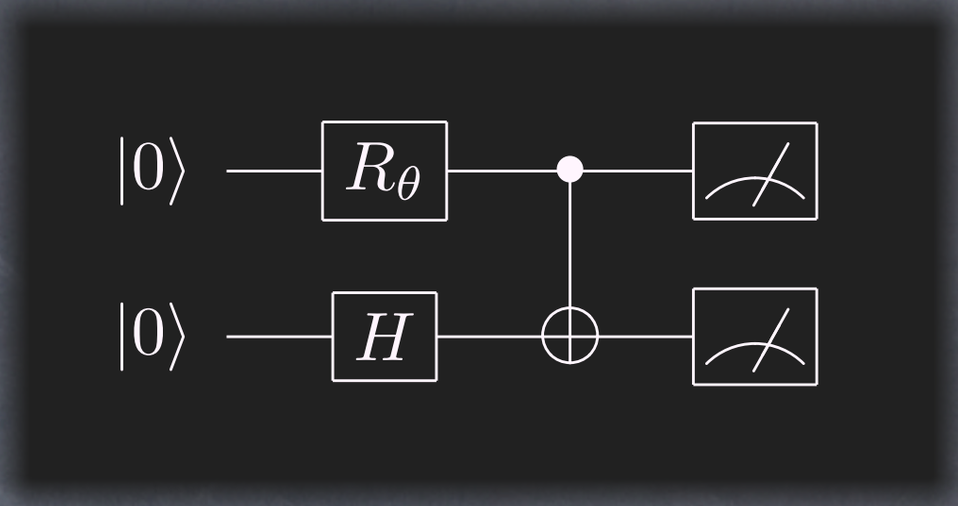
# Measurement Based QC






[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

# Measurement Based QC

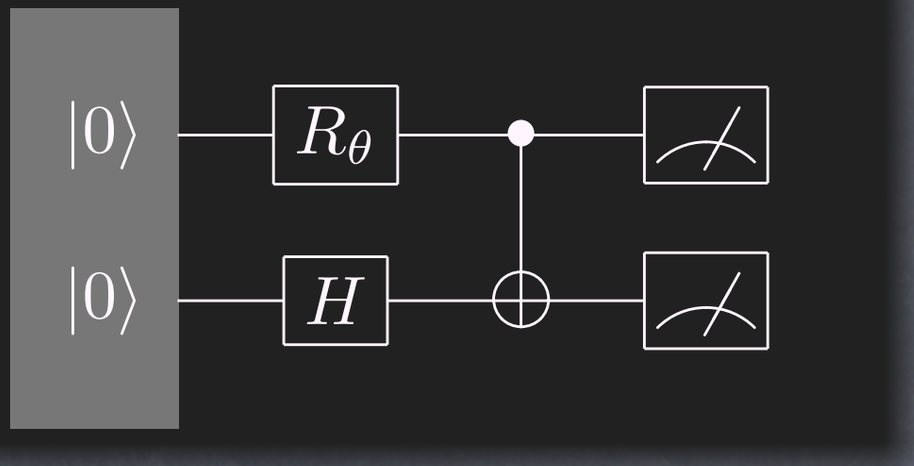





Create Entangled State

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

# Measurement Based QC



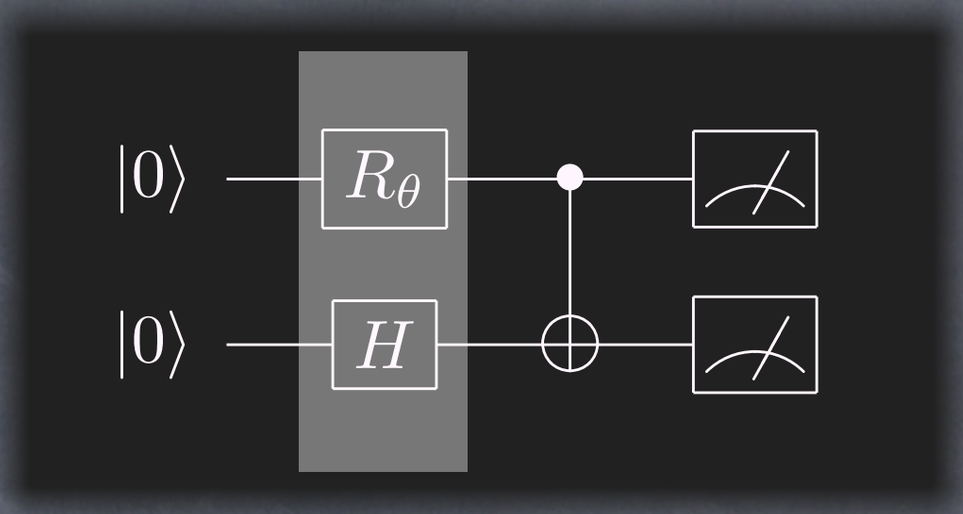




Adaptively measure to enact circuit

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

# Measurement Based QC



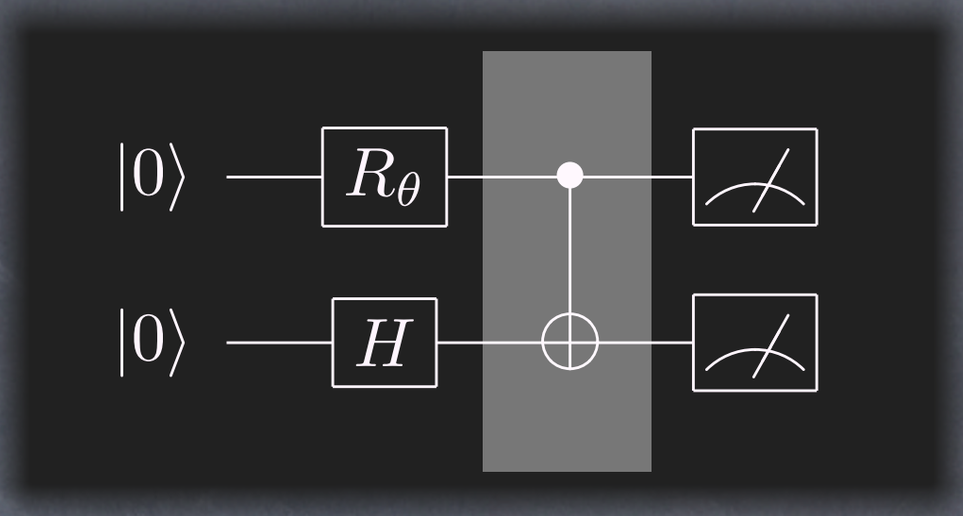


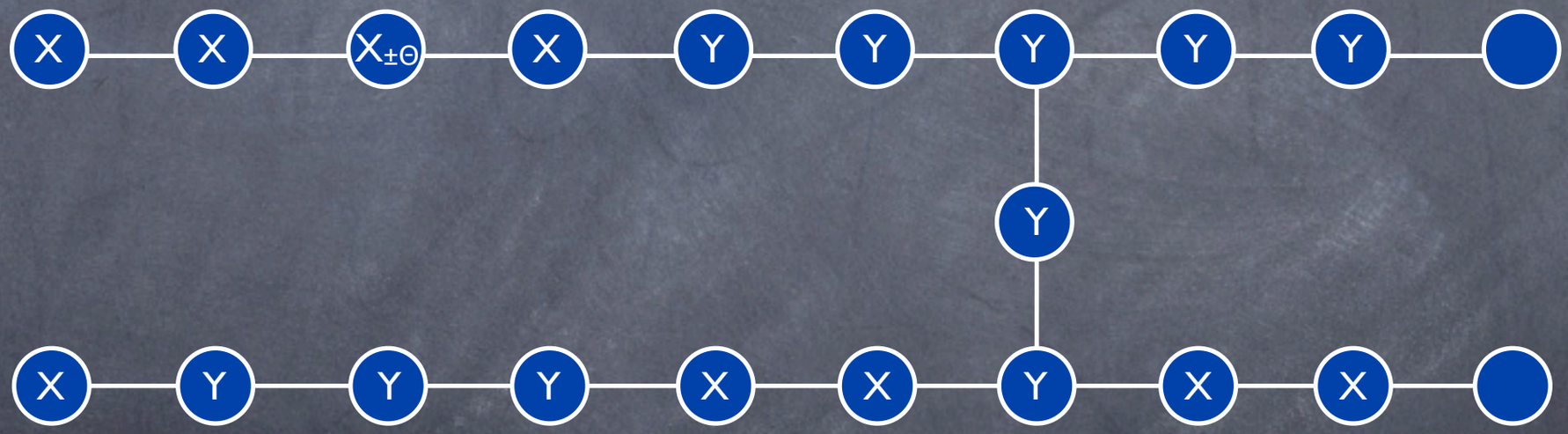


Adaptively measure to enact circuit

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

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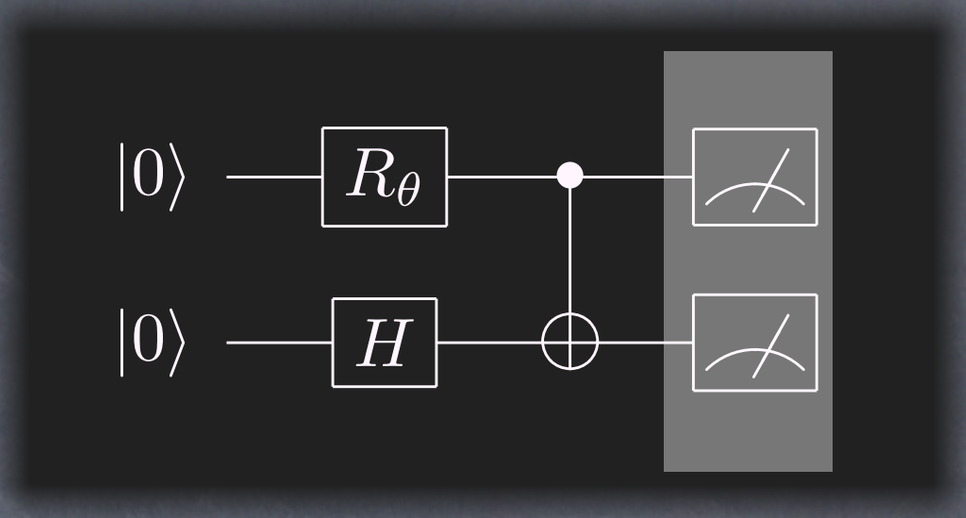


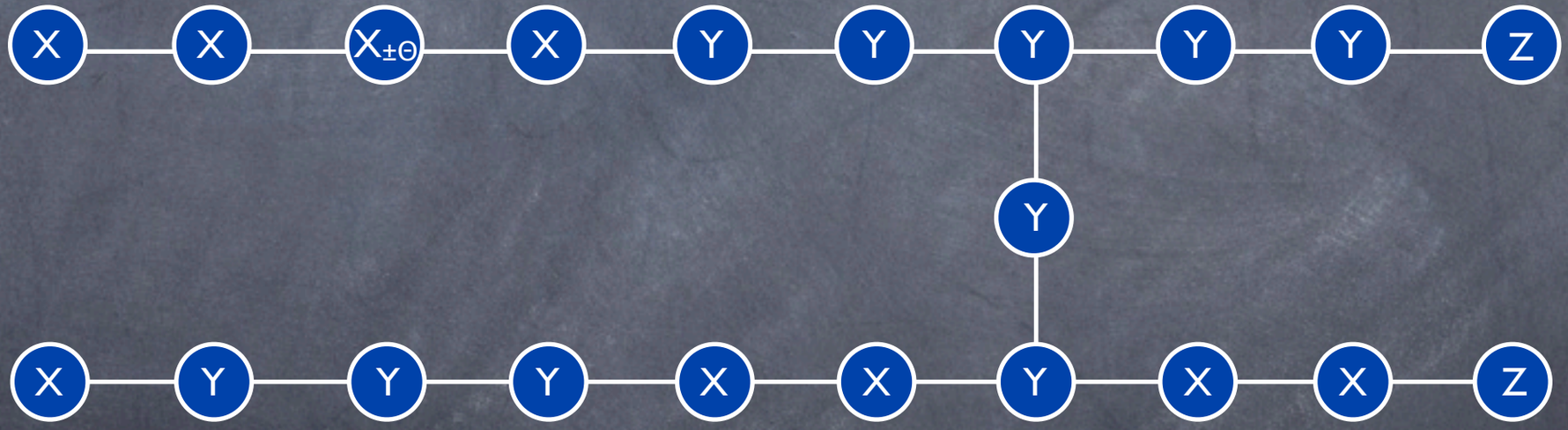




Adaptively measure to enact circuit

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

# Measurement Based QC



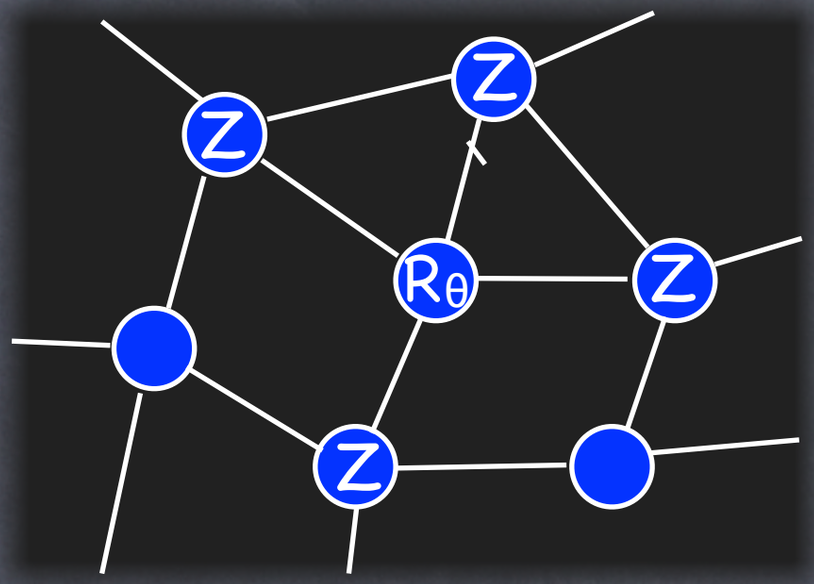




Adaptively measure to enact circuit

Initial state described locally + measurements are all local

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

# Twisted Graph State




$$= [R_\theta]_v \prod_{w|(v,w)\in\mathcal{E}} [Z]_w$$

$$R_\theta = \cos\theta X + \sin\theta Y$$

"Vertex operator with angle θ"

Twisted Graph State: is common +1 eigenstate of all vertex operators (one per vertex)

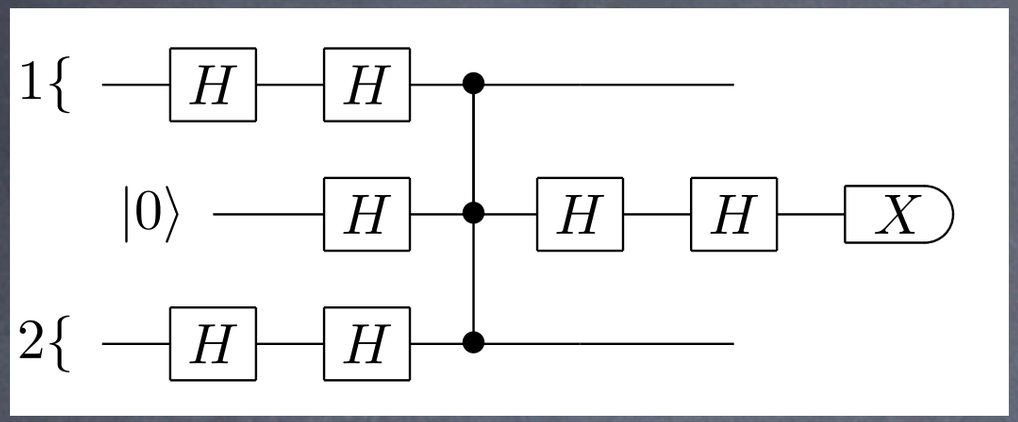
# QC to Twist Graph State

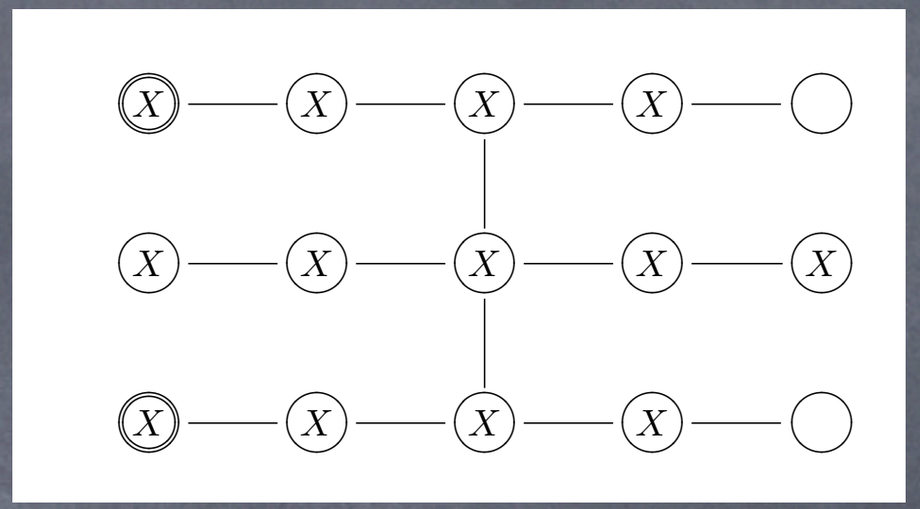
| Circuit Element | Graph Gadget |
|---|---|
| $\|0\rangle$ — | $X$ — $X$ — |
| input | $X$ — |
| internal | — $X$ — $X$ — |
| output | — $X$ — ◯ |
| $H$ — $R(\theta)$ — | — $X^{\theta}$ — |
| $H$ — $R(\theta)$ ● <br> $H$ — $R(\phi)$ ● | $X^{\theta}$ <br> $X^{\phi}$ |
| $H$ — $R(\theta_1)$ ● <br> $H$ — $R(\theta_2)$ ● <br> ⋮ ⋮ ⋮ <br> $H$ — $R(\theta_{m-1})$ ● <br> $H$ — $R(\theta_m)$ ● | $X^{\theta_1}$ <br> $X^{\theta_2}$ <br> ⋮ <br> $X^{\theta_{m-1}}$ <br> $X^{\theta_m}$ |
| $H$ — $X$ ) | — $X$ |
| — $X$ ) | — $X$ — $X$ |

Quantum circuit made up of preparations, gates, and measurements

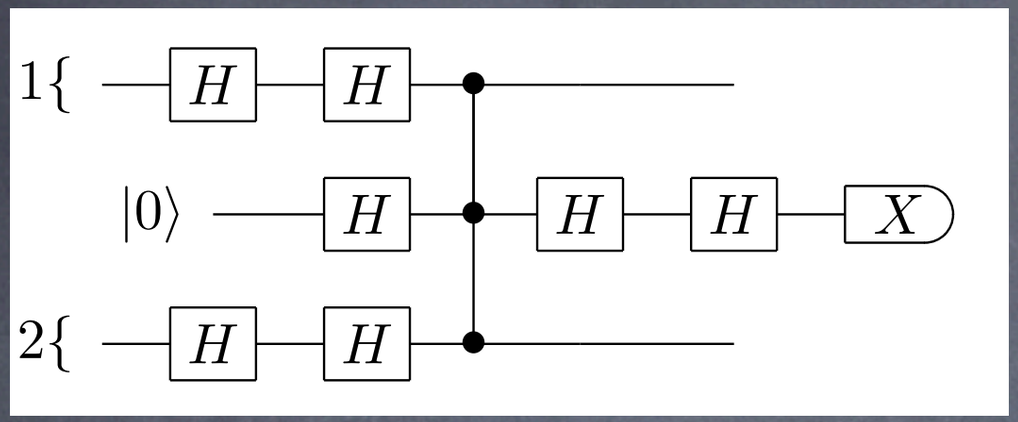
Graph describing vertex operators and places where X measurements would be made to enact MBQC for the quantum circuit

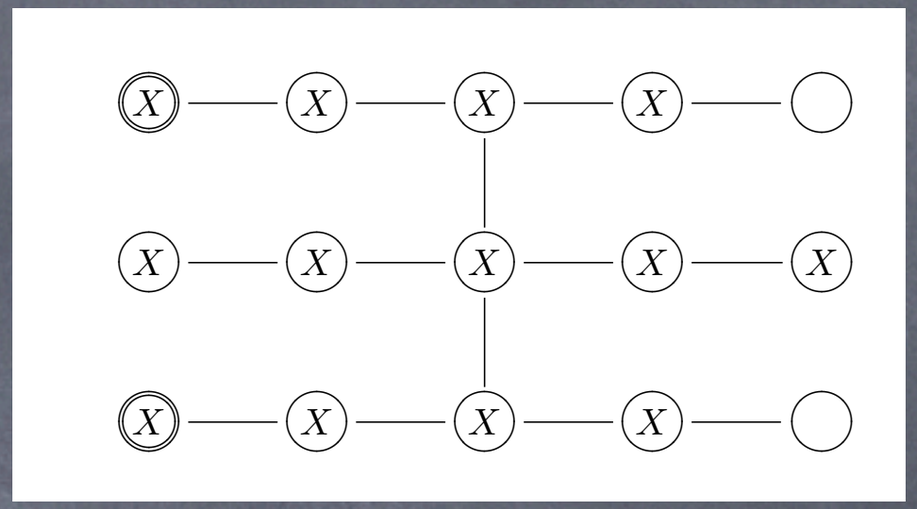
# Encoded MBQC Example



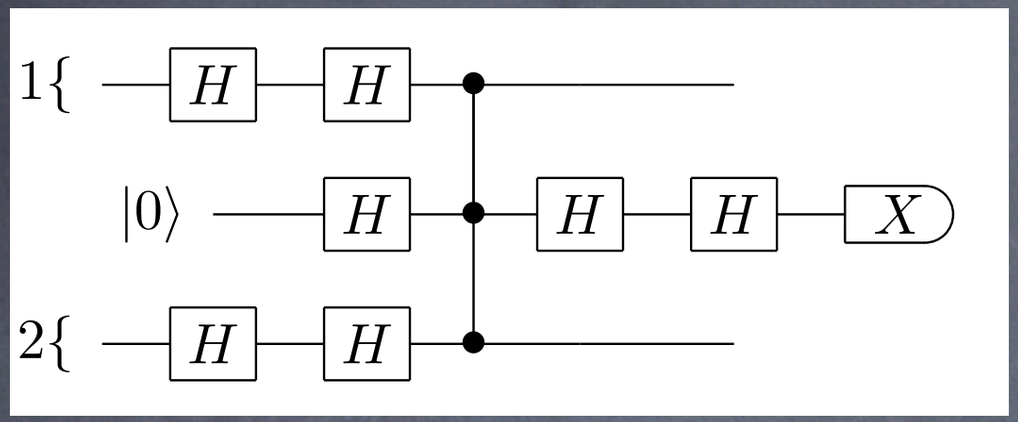
# Encoded MBQC Example

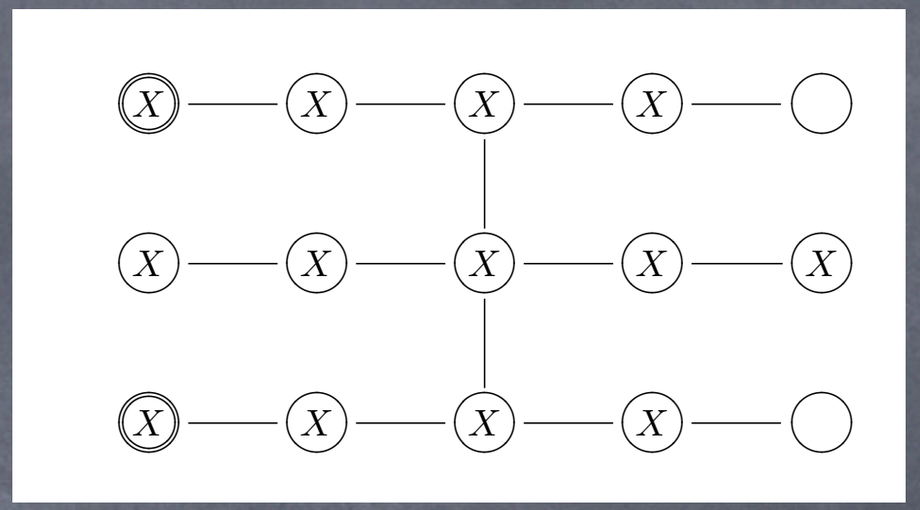



1. vertex operator for every non-double-circled node

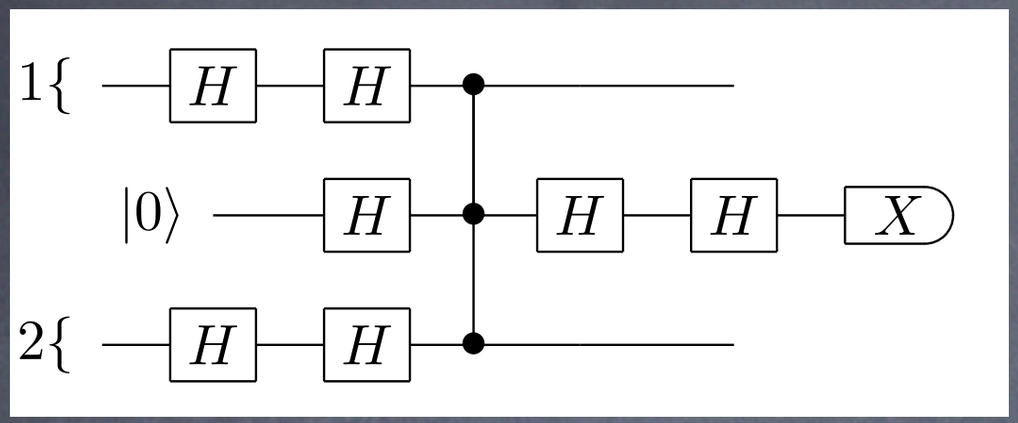
# Encoded MBQC Example

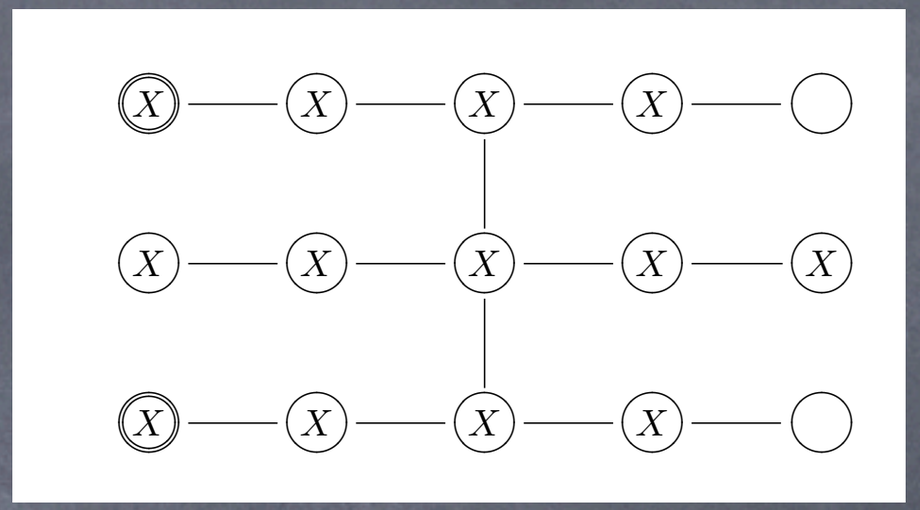



1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators

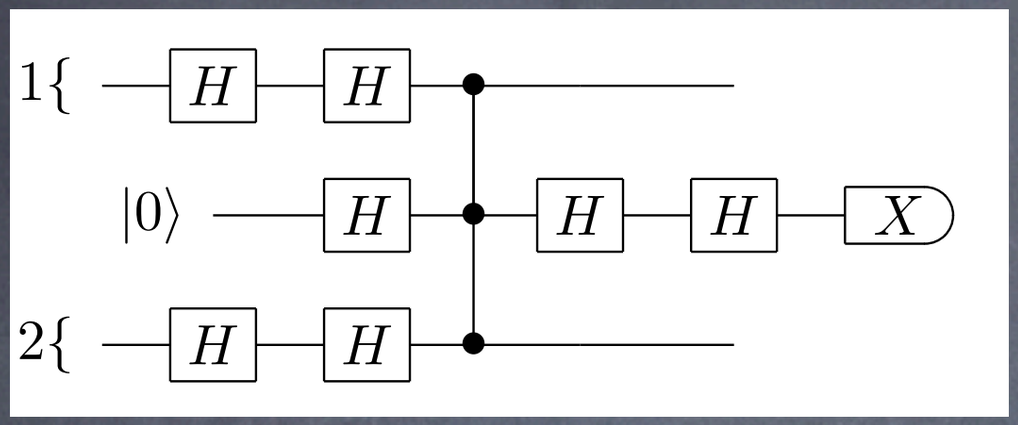
# Encoded MBQC Example

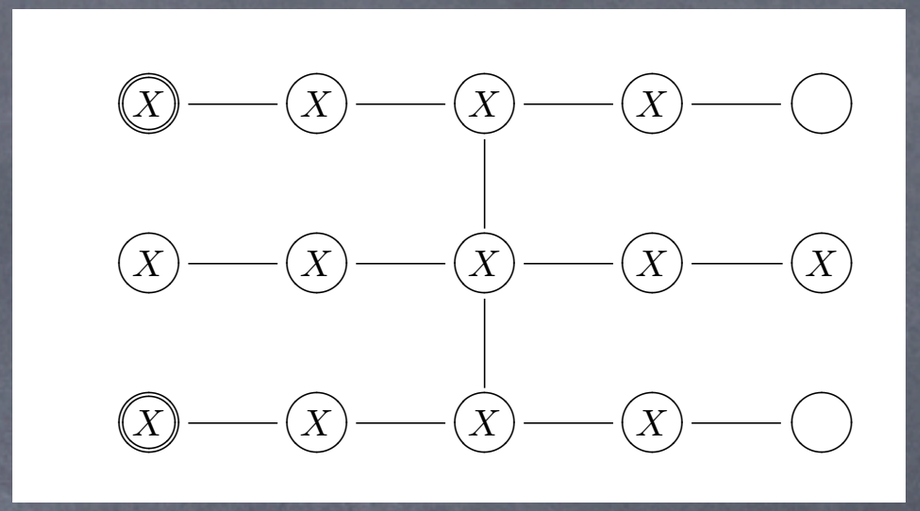



1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators
3. input information encoded @ double-circled nodes

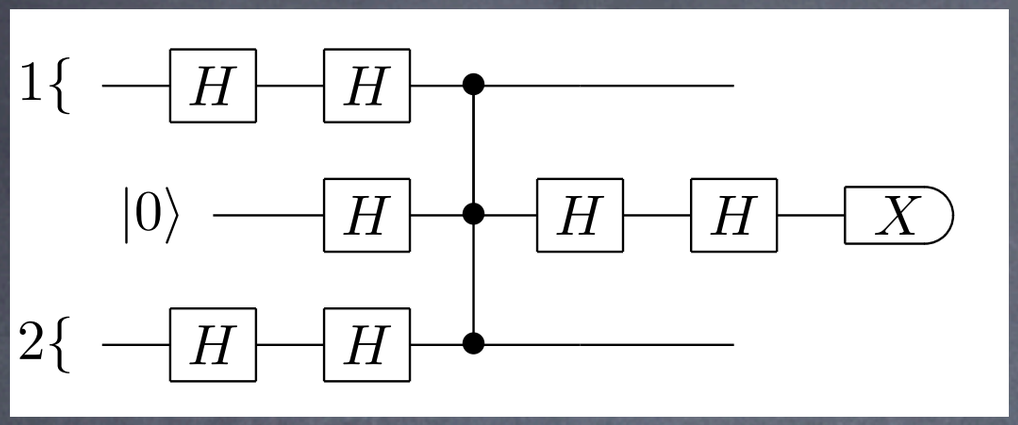
# Encoded MBQC Example

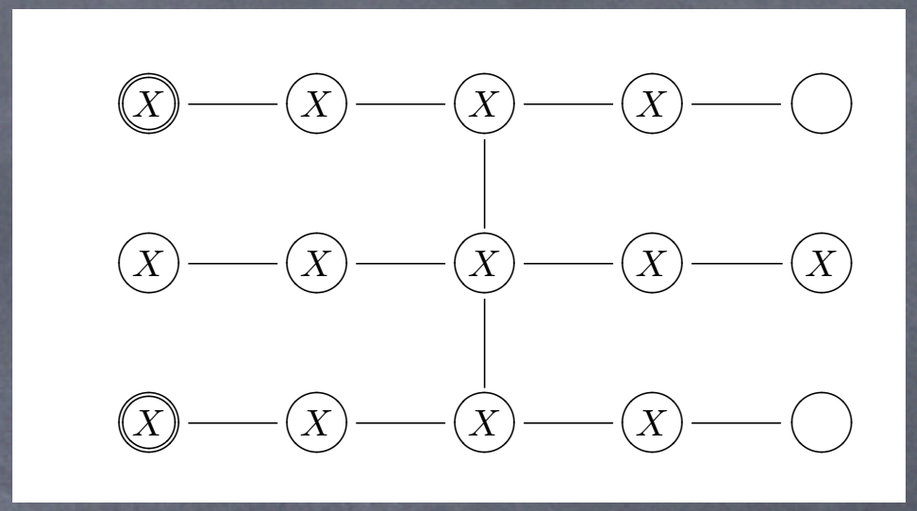



1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators
3. input information encoded @ double-circled nodes
4. X measurements occur on all vertices labeled X

# Encoded MBQC Example



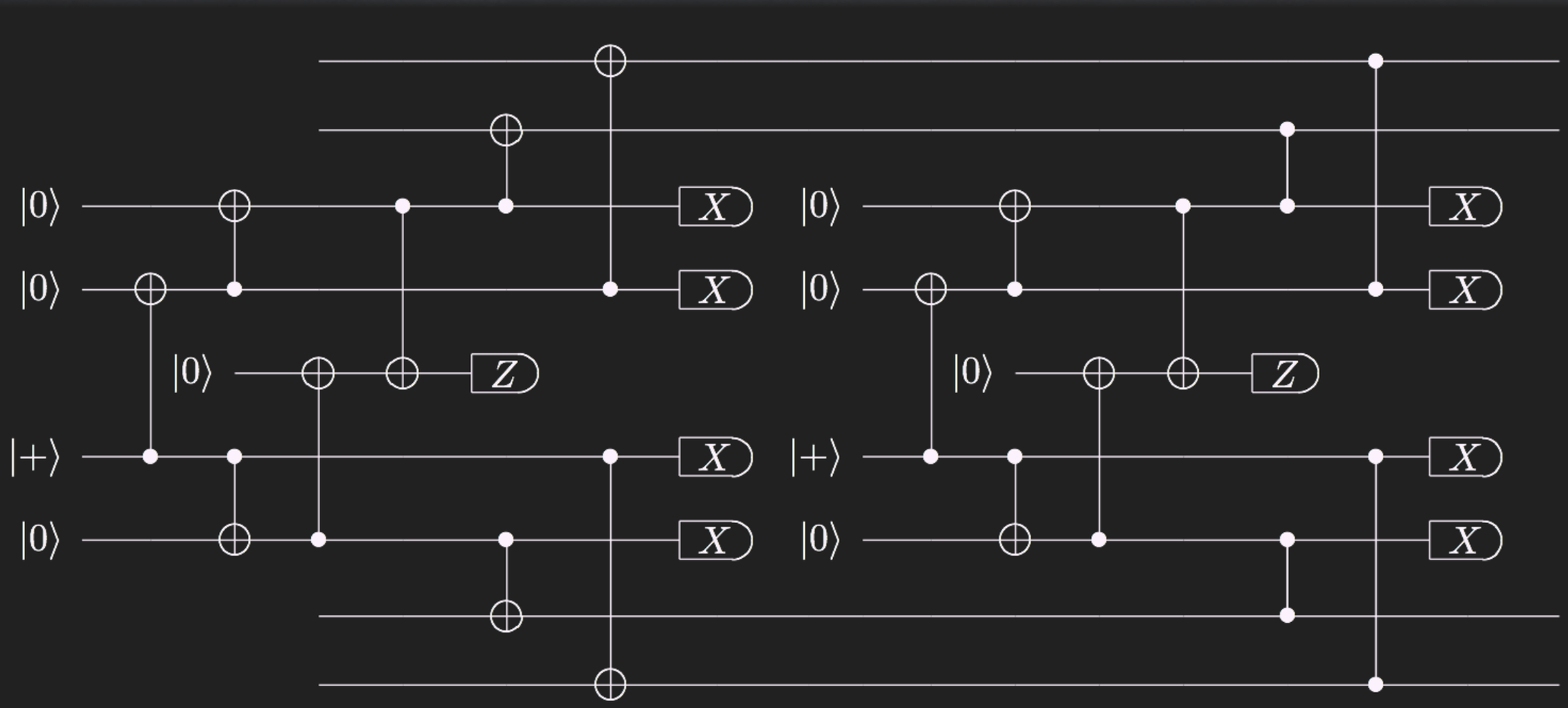




1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators
3. input information encoded @ double-circled nodes
4. X measurements occur on all vertices labeled X
5. Output measurements at X's, unmeasured output qubits at non-X-ed nodes

# CONSTRUCTION

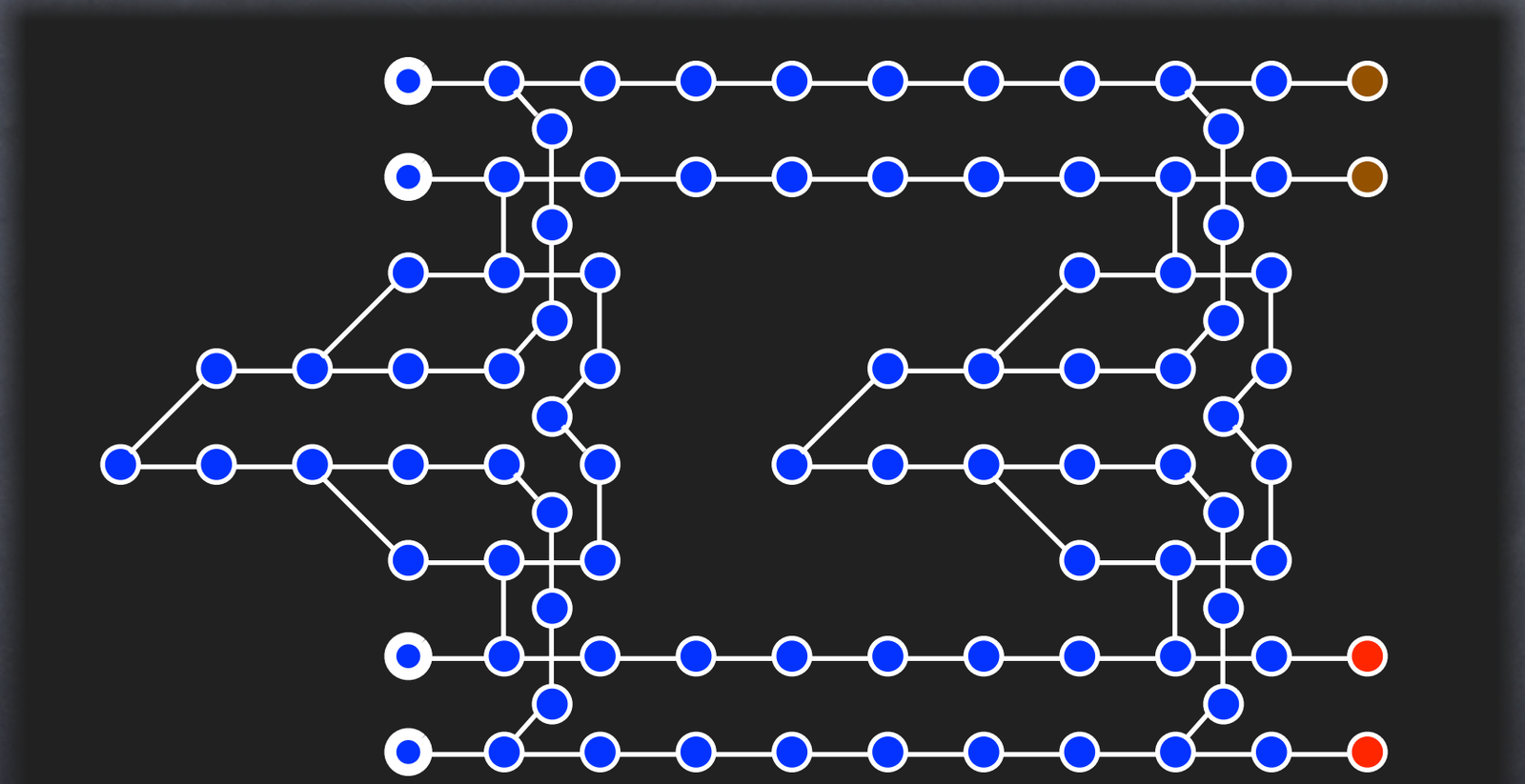
Input: [[n,k,d]] stabilizer code

1. Construct fault-tolerant syndrome measuring quantum circuit that is k-neighboring in space-time on lattice in dimension D (D>1)



# CONSTRUCTION

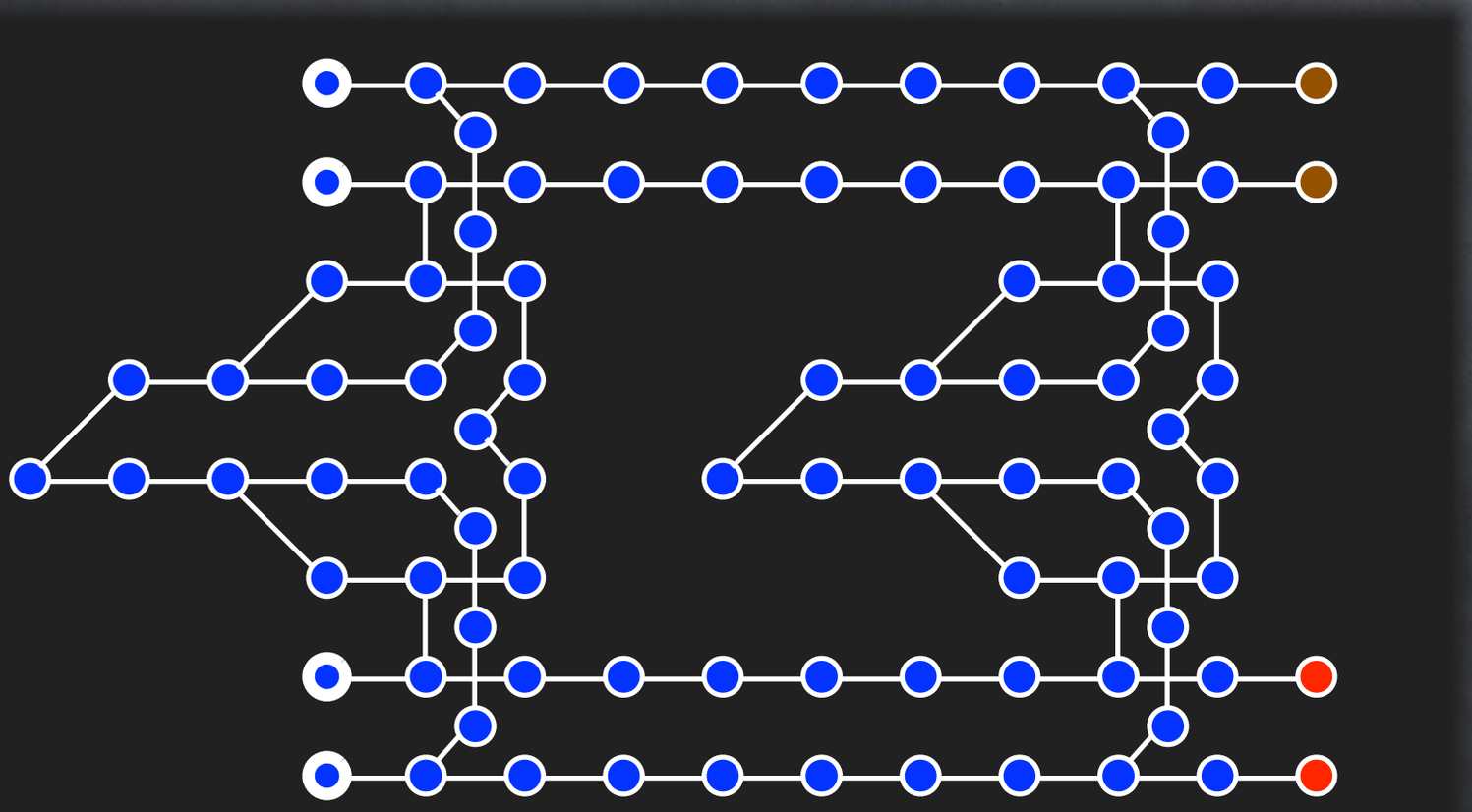
2. Convert this circuit to the scheme that would be used to implement it in MBQC



[Related: Rausendorf, Harrington, Goyal 2006]

# CONSTRUCTION

3. Take operators from MBQC construction (vertex operators and X operators for measurements) and identify (a) stabilizer subgroup of these operators and (b) gauge qubits



$X^{\otimes 4}$ and $Z^{\otimes 4}$ stabilizer localized on red qubits

# Profit

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using t-neighboring measurements (t constant).

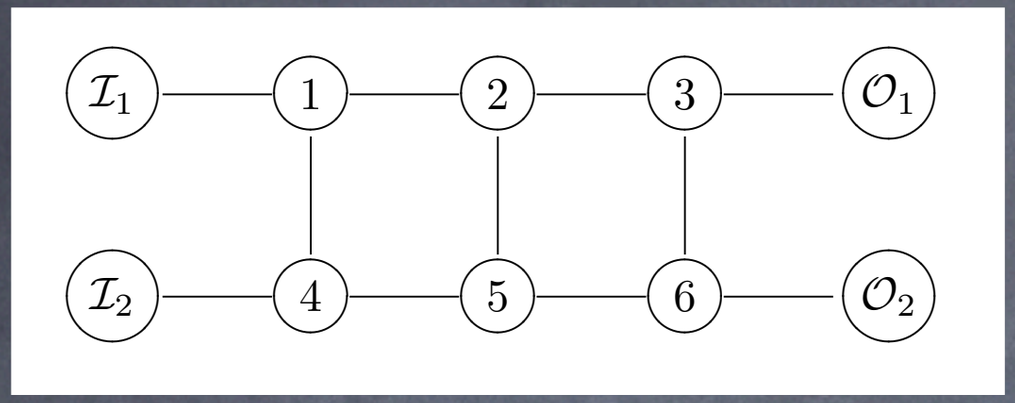
$$$$

# FAQ

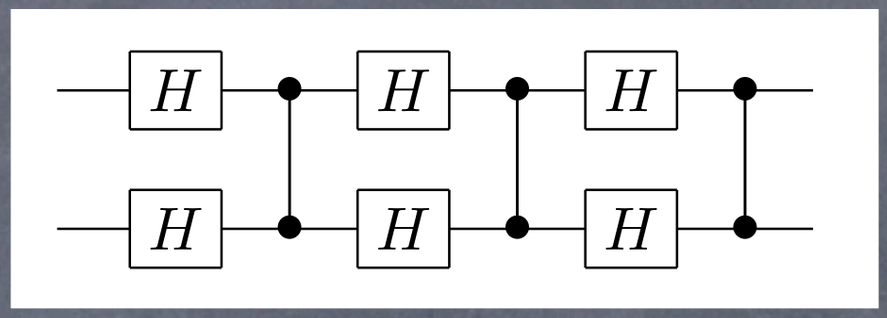
Q: Why fault-tolerant syndrome measuring circuits?

- These maintain the distance of the code, actually non-fault-tolerant circuits are possible as well.

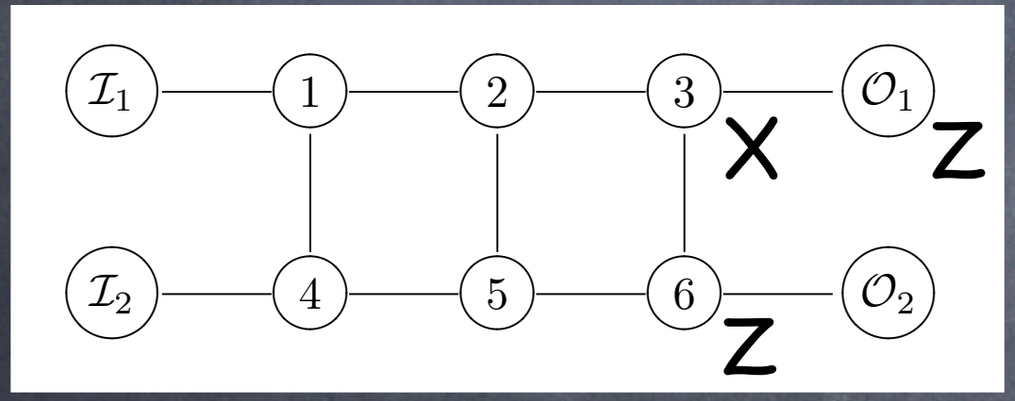
# An Obstacle



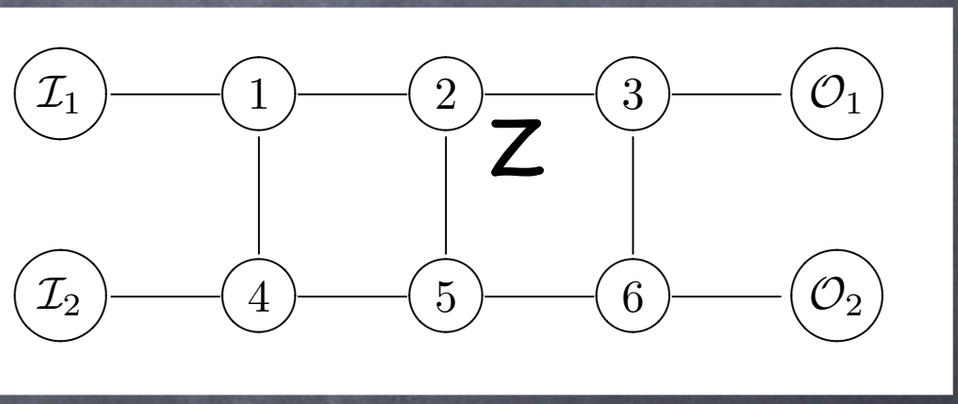

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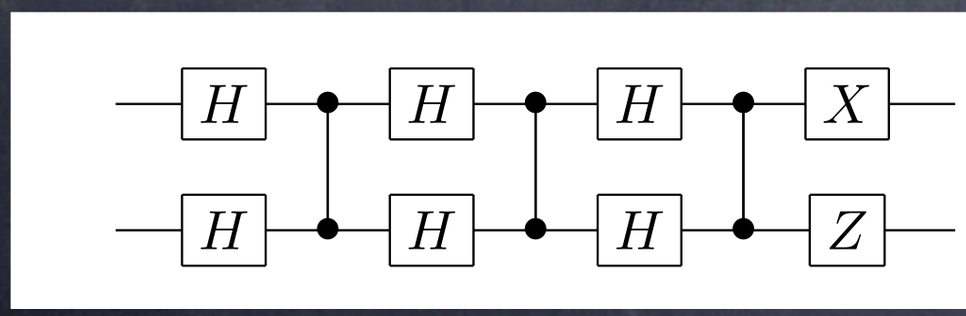



swap

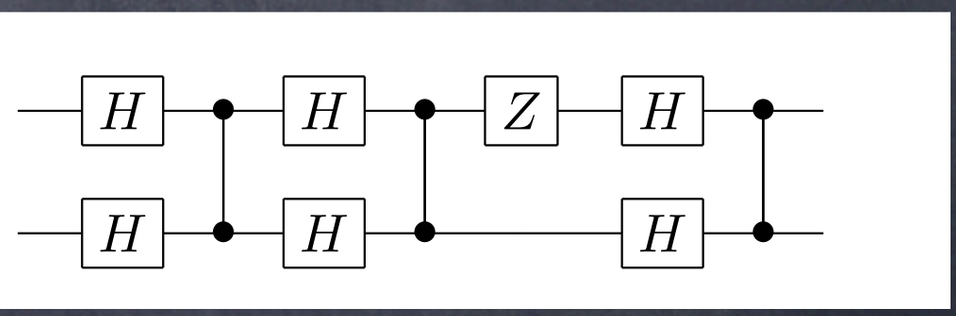



equiv




equiv



# FAQ

Q: Why fault-tolerant syndrome measuring circuits?

- These maintain the distance of the code, actually non-fault-tolerant circuits are possible as well.

Q: But aren't fault-tolerant circuits probabilistic?

- Often yes. Use the circuit as if the measurement result turned out in your favor

Q: Do these codes have memory thresholds?

- Yes. Basic idea is that error correction measurements can be done in constant depth, then leverage FT threshold.

Q: k-local for what k?

- Can be made 3-local

# Consequences




arXiv:soon

# QuESTs

Self correction:

Do these Hamiltonian made of gauge qubits on these codes self-correct?

$$H = \sum_{g \in \mathcal{G}} G$$

Ancilla terms cost energy for errors.

FT Universal Adiabatic QC:

These codes are directly related to adiabatic quantum transistors and provide ideas for how to exactly diagonalize variations on these models

# Review

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using t-neighboring measurements (t constant).

# Q-Dub Group

## Grads

**Greg Crosswhite (Physics)**
Isaac Crosson (Physics)
Lukas Svec (Physics)
Jijiang Yan (Physics)
Kamil Michnicki (Physics)
David Rosenbaum (CS)
Paul Pham (CS)
Kevin Zatloukal (CS)
Punya Biswall (CS)

## Faculty

Dave Bacon
**Aram Harrow**

## Undergrads

Zakk Webb (Physics/CS/Math)
Kate Liotta (Physics/CS)
**Jonathan Shi (Physics/CS)**
Melanie Jensenworth (Math/CS)
Jeffrey Booth (CS)
Rob McClure (CS)
Harshad Petwe (CS)

# The Successful Test Dog



No cats were hurt during this talk, but they should have been